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Beam Propagation Simulation in SiO_2

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Contents

1 Introduction

Since their invention in the early 80's [2], femtosecond lasers have rapidly advanced the frontiers of materials science research due to their high peak intensity and ultra short temporal resolution.

The ultra-short nature of femtosecond pulses has enabled imaging of ultrafast processes, such as chemical reactions [3] and the complex energy transfer dynamics between electrons and crystal lattices in solids [4], whereas the high peak intensity of these pulses allows them to ionise matter and trigger laser-produced plasma (LPP), which emits a variety of spectra, including X-rays [5]. Among these nonlinear effects, Kerr self-focusing, plasma defocusing, and multiphoton/avalanche ionization play a critical role in the dynamics of femtosecond pulse propagation in dielectrics.

Studying the dynamics of femtosecond pulses in dielectric media, ranging from the near-UV to the near-infrared, helps clarify the transfer of electromagnetic energy to electrons, followed by electron-lattice interactions that alter the lattice and convert energy into heat [5]. However, such studies remain challenging, and some phenomena remain difficult to model for wide-bandgap dielectric materials such as silicon dioxide (SiO_2). In this study, we focus on SiO_2, a material widely used in optics due to its transparency and nonlinear properties.

Indeed, these media are transparent under low-intensity irradiation, and when they are irradiated by an intense femtosecond pulse, a cascade of effects begins, ranging from nonlinear ionisation occurring within a few femtoseconds to thermal effects occurring on picosecond to nanosecond timescales [7]. One of the most striking phenomena in this context is filamentation, where the beam undergoes self-focusing due to the Kerr effect, balanced by plasma defocusing, leading to the formation of stable light filaments. Once nonlinear ionisation becomes significant, the material loses its transparency, leading to photo-induced damage.

Femtosecond laser processing techniques have an edge over continuous-wave laser machining techniques because of the extremely fast interaction time between the beam and the sample. The damage induced by femtosecond laser pulses is mainly driven by electronic excitation and associated nonlinear processes before the material lattice has equilibrated with the excited carriers [5]. This allows for better control over collateral thermal damage and thermal stresses, which are typically associated with continuous laser processing. Also, the nonlinear response allows for structural modifications of dielectrics inside the bulk and paves the way for microstructuring that creates subwavelength "voxels" [5].

Theoretical models for femtosecond pulse propagation in dielectrics have been extensively developed [9]. The complexity of these nonlinear interactions has pushed for extensive research aimed at understanding their fundamental mechanisms and harnessing these properties to develop precision laser processing techniques.

This work presents a simulation of beam propagation in SiO_2 using the nonlinear Schrödinger equation and equations governing electron density evolution. We will explain the physics underlying the model, as well as some of the numerical methods used to solve this complex system.

2 Excitation of Free Carriers

Current experimental methods for generating femtosecond lasers in the near ultraviolet range (NUV) remain complex, which limits research on the linear optical transitions of wide bandgap dielectric materials such as SiO_2.

Given these experimental limitations, research has focused mainly on the study of nonlinear absorption processes that generate free carriers in the conduction band. By responding to the electric field, these electrons increase their energy and trigger effects such as avalanche ionisation, which modify the optical properties of the material, in particular its absorbance [5, 7].

In this section, we explain the nonlinear mechanisms responsible for the generation of free carriers, that is, the excitation of electrons from the valence band to the conduction band. In wide bandgap dielectrics, these carriers are central to the light matter interaction, since they are the only way to transfer optical energy to the crystal lattice of the material, which is otherwise transparent to low intensity light.

2.1 Photoionization

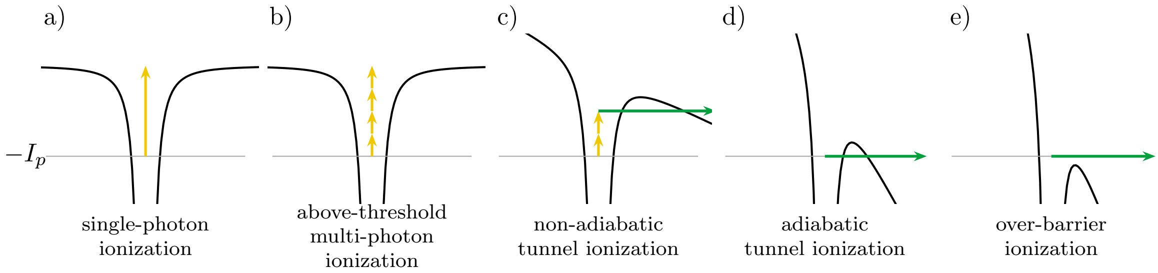


Figure 1: Illustration of the different ionization regimes.

Strong focusing leads to a high optical intensity in the material. This strong concentration of photons at the same location increases the probability of multiphoton ionization, a nonlinear effect [5]. The more intense the field, the more the potential binding the electron to the ion is deformed. Beyond a certain intensity threshold, the barrier takes a triangular shape and becomes thin enough to allow tunnel ionization. The Keldysh parameter was introduced to distinguish between the regime dominated by multiphoton ionization and the regime dominated by tunnel ionization [5].

In the semiclassical model, an electron escapes through a barrier of width $l = U_i/e\mathcal{E}$ with a velocity $v = \sqrt{2U_i/m}$. The time needed to cross this distance is:

$$\tau_{\text{tun}} = \frac{l}{v} = \frac{U_i/e\mathcal{E}}{\sqrt{2U_i/m}} = \frac{\sqrt{2mU_i}}{e\mathcal{E}} \tag{1}$$

The characteristic time during which tunnel ionization can occur is given by $\tau \approx 1/\omega_0$. The ratio of these two times is then:

$$\gamma = \frac{\tau_{\text{tun}}}{\tau} = \omega_0 \frac{\sqrt{2mU_i}}{e\mathcal{E}} \tag{2}$$

If the ratio $\gamma \ll 1$, the tunnel crossing time is very short compared to the ionization time. Tunnel ionization is then instantaneous on the scale of the laser cycle, and the field behaves as a quasi-static field. Conversely, if $\gamma \gg 1$, the field oscillates too fast for the electron to cross the barrier by tunneling, and ionization then requires multiphoton absorption [7].

The Keldysh photoionization rate is defined as follows:

$$W_{PI}(|\mathcal{E}|) = \frac{2\omega_0}{9\pi}\left(\frac{\omega_0 m}{\hbar\sqrt{\gamma_1}}\right)^{\frac{3}{2}} Q(\gamma, x)\exp(-\alpha\lceil x+1\rceil) \tag{3}$$

With $\gamma_1 = \frac{\gamma^2}{1+\gamma^2}$, $\gamma_2 = \frac{1}{1+\gamma^2}$ and $x = \frac{2}{\pi}\frac{U_i E(\gamma_2)}{\hbar\omega_0\sqrt{\gamma_1}}$

$$Q(\gamma, x) = \frac{\pi}{2K(\gamma_2)}\sum_{n=0}^{\infty}\exp\left(-n\pi\frac{K(\gamma_1)-E(\gamma_1)}{E(\gamma_2)}\right)\Phi\left(\frac{\pi}{2}\sqrt{\frac{n+2\lceil x+1\rceil-2x}{K(\gamma_2)E(\gamma_2)}}\right) \tag{4}$$

Φ is the Dawson function $\Phi(z) = e^{-z^2} \int_0^z e^{y^2} dy$ and K and E are respectively the complete elliptic integrals of the first and second kind, with the parameter $m = k^2$

$$K(m) = \int_0^{\pi/2} [1 - m \sin(t)^2]^{-1/2} dt \tag{5}$$

$$E(m) = \int_0^{\pi/2} [1 - m \sin(t)^2]^{1/2} dt \tag{6}$$

As observed in reference [8], filamentation in fused silica typically occurs at intensities around 5×10^{13} W/cm^2, where multiphoton and tunnel ionization both contribute significantly.

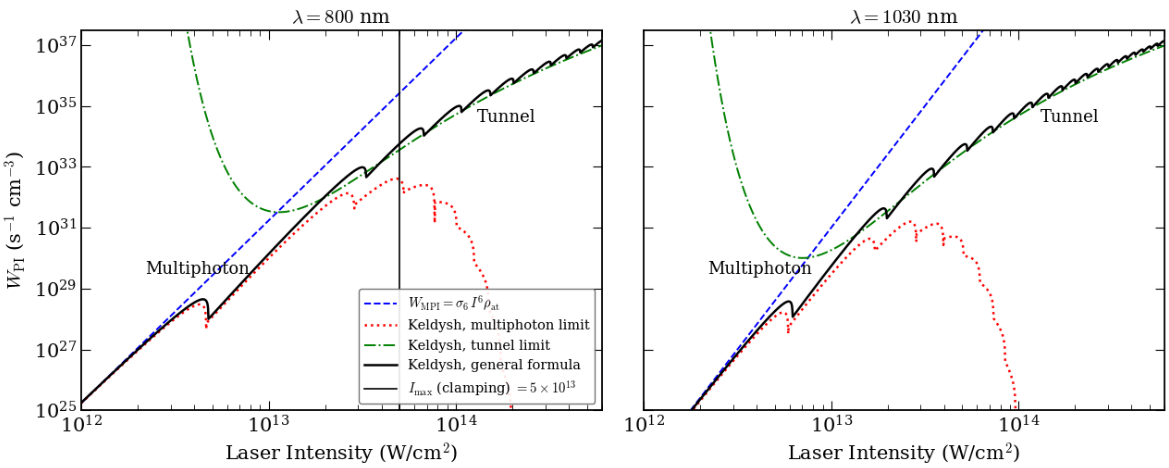


Figure 2: The photoionization rate W_{PI} in fused silica ($U_i = 9$ eV) is plotted against laser intensity for $\lambda = 800$ nm and $\lambda = 1030$ nm. The solid black line represents the full general Keldysh formula, while the dotted red and dash dotted green lines show its exact analytical asymptotes for the multiphoton limit ($\gamma \gg 1$) and the tunnel limit ($\gamma \ll 1$) respectively. The dotted blue line corresponds to the standard perturbative MPI law ($W_{\mathrm{MPI}} = \sigma_K I^K \rho_{\mathrm{at}}$). The vertical gray line indicates the typical clamping intensity level ($\sim 5 \times 10^{13}$ W/cm^2) reached in filamentation simulations at 800 nm.

2.2 Free Carrier Absorption

Once electrons are promoted to the conduction band, they behave as quasi free carriers. The laser field accelerates them, and they drift in opposition to the electrical field $\vec{E}$ [5]. One might expect these carriers to keep absorbing photons and gaining energy directly. However, single photon absorption by an isolated free electron is forbidden, because it cannot simultaneously satisfy conservation of energy and conservation of momentum.

For the absorption of a photon $(\omega_0, \vec{k}_{\mathrm{ph}})$ by an electron with initial wave vector $\vec{k}_i$ moving to a final state with wave vector $\vec{k}_f$, the laws of conservation of energy and momentum must be satisfied. They read:

$$E_f = E_i + \hbar\omega_0, \qquad \vec{k}_f = \vec{k}_i + \vec{k}_{\mathrm{ph}}. \tag{7}$$

Using the parabolic band $E = \hbar^2 k^2 / 2m^*$ and the photon momentum $k_{\mathrm{ph}} = \omega_0 / c = 2\pi/\lambda_0$, equation (7) becomes:

$$\frac{\hbar^2 k_f^2}{2m^*} = \frac{\hbar^2 k_i^2}{2m^*} + \hbar\omega_0, \qquad k_f = k_i + \frac{2\pi}{\lambda_0}. \tag{8}$$

The photon momentum is set by the wavelength alone. At $\lambda_0 = 1030$ nm, $k_{\text{ph}} = 2\pi/\lambda_0 \approx 6.1 \times 10^6$ m^{-1}. The electron momentum, on the other hand, is set by its kinetic energy in the conduction band. Taking a representative value $E_i \sim 1$ eV, and any energy of order one eV leads to the same conclusion, we obtain:

$$k_i = \frac{\sqrt{2m^* E_i}}{\hbar} \approx 5 \times 10^9 \text{ m}^{-1}, \qquad \frac{k_{\text{ph}}}{k_i} \approx 10^{-3}. \tag{9}$$

Conservation of momentum therefore forces $k_f \approx k_i$, hence $E_f \approx E_i$, which contradicts $E_f = E_i + \hbar\omega_0$ unless $\hbar\omega_0 \approx 0$.

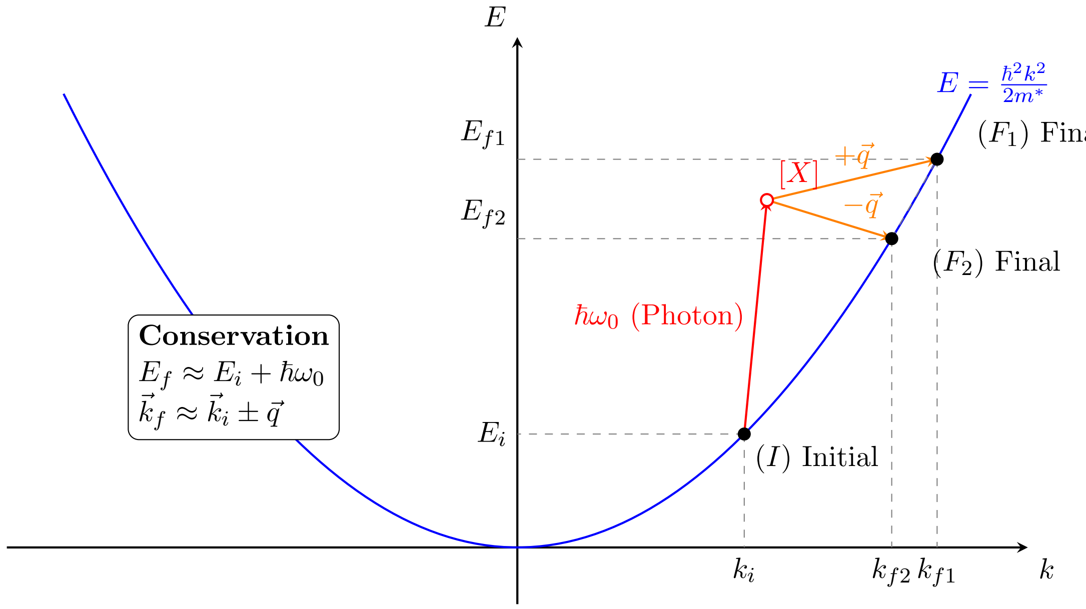
The process becomes allowed if a third body carries away the momentum mismatch. In a polar dielectric such as SiO_2, this third body is a phonon of wave vector $\vec{q}$ and energy $\hbar\omega_q$, so that:

$$E_f = E_i + \hbar\omega_0 \pm \hbar\omega_q, \qquad \vec{k}_f = \vec{k}_i + \vec{k}_{\text{ph}} \pm \vec{q}. \tag{10}$$

Since phonon wave vectors span the entire Brillouin zone:

$$q \lesssim \tfrac{\pi}{a} \sim 10^{10} \text{ m}^{-1} \gtrsim k_i$$

Momentum can now be balanced, while $\hbar\omega_q \sim 0.06$–0.15 eV $\ll \hbar\omega_0$ leaves the energy balance essentially unchanged. This phonon assisted mechanism is free carrier absorption (FCA), equivalently inverse Bremsstrahlung [5, 7].



2.3 Avalanche Ionization

The energy shift is given by $\Delta E = \hbar\omega_0 \pm \hbar\omega_q$, where $\hbar\omega_q$ is the phonon energy and $\hbar\omega_0$ is the photon energy. Since the phonon energy is very small compared to the photon energy, we can assume $\Delta E \approx \hbar\omega_0$. Electrons can absorb several photons in sequence, increasing their velocity and their energy. Beyond a certain threshold, the energy of the free carriers exceeds the gap energy, and they can then collide with electrons of the valence band, causing impact ionization. This leaves two electrons in the conduction band, which increases the absorption efficiency and triggers further impact ionization events. This cascading effect is known as avalanche ionization. It combines free carrier absorption with impact ionization, sketched in Figure 3.

This process can repeat itself as long as the electric field is present and intense enough.

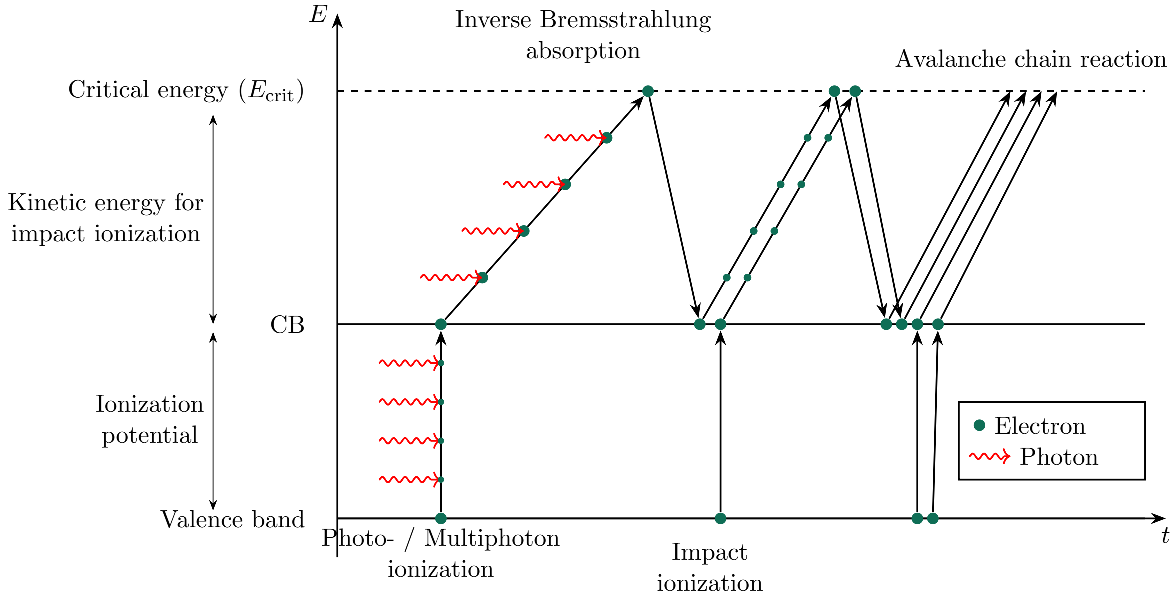


Figure 3: Avalanche chain reaction. A seed electron is promoted to the conduction band by photoionization, heated above the critical energy E_{crit} by successive inverse Bremsstrahlung absorption, and knocks a second valence electron into the conduction band by impact ionization. Both electrons repeat the cycle, doubling the population at each generation.

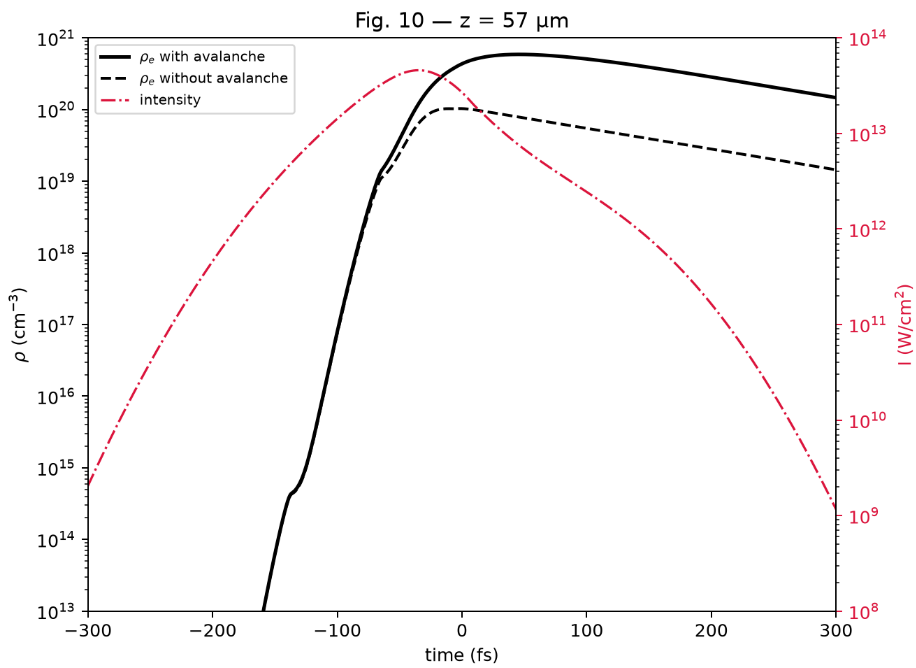


Figure 4: On-axis electron density at fixed $z = 57\ \mu$m with and without the avalanche term $\sigma_\omega IN/U_i$ of Eq. (35), reproducing Fig. 10 of [8] at 1.1 μJ, 800 nm. Photoionization alone (dashed) already produces a plateau once the pulse has passed, but avalanche multiplication (solid) drives the final density roughly a factor of six higher, showing that impact ionization contributes substantially to the total carrier yield once seeded by the multiphoton channel, even for the ~ 100 fs pulse duration used here.

2.4 Optical Kerr Effect, Self-Phase Modulation and Self-Steepening

When a laser beam travels through a dielectric material, it induces a microscopic displacement of charges, forming oscillating electric dipoles that combine to produce a macroscopic polarization. This leads to a change in the refractive index. When the electric field is intense enough, the response becomes nonlinear, giving rise to new propagation effects such as self-focusing, self-phase modulation, and plasma defocusing.

To derive the nonlinear polarization and the resulting Kerr effect in the SI system, we start from the definition of the electric displacement field:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \tag{11}$$

where $\mathbf{P}$ can be expanded to include the contribution of the third order susceptibility. Applied to a monochromatic electric field $\mathbf{E} = \mathcal{E} e^{i(k_0 z - \omega_0 t)} + \text{c.c.}$, this expression becomes:

$$\mathbf{P} = \varepsilon_0 \left(\chi^{(1)} \mathbf{E} + \chi^{(3)} \mathbf{E}^3 \right) \tag{12}$$

Expanding $\mathbf{E}^3$ and neglecting the $3\omega_0$ terms associated with third harmonic generation (THG), we isolate the term oscillating at the fundamental frequency, which represents the nonlinear response of the medium at the angular frequency ω_0:

$$\mathbf{P} \approx \varepsilon_0 \left[\chi^{(1)} + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right] \mathbf{E} \tag{13}$$

Substituting this expression into the displacement field equation, we obtain:

$$\mathbf{D} = \varepsilon_0 \left(1 + \chi^{(1)} + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right) \mathbf{E} = \varepsilon_0 \left(n_0^2 + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right) \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} \tag{14}$$

where $n_0^2 = 1 + \chi^{(1)}$. The effective refractive index n is defined as the square root of the relative permittivity, $n = \sqrt{\varepsilon_r} = \sqrt{n_0^2 + \frac{3}{4}\chi^{(3)}|\mathcal{E}|^2}$. For weak nonlinearities, a first order Taylor expansion gives:

$$n \approx n_0 + \frac{3\chi^{(3)}}{8n_0} |\mathcal{E}|^2 \tag{15}$$

Comparing this expression with the standard form $n = n_0 + n_2 I$, where the intensity is $I = \frac{1}{2} n_0 c \varepsilon_0 |\mathcal{E}|^2$, we identify the nonlinear refractive index coefficient:

$$n_2 = \frac{3\chi^{(3)}}{4 n_0^2 c \varepsilon_0} \tag{16}$$

This refractive index therefore depends on intensity. This is the origin of the optical Kerr effect. Since the intensity $I(r)$ of a Gaussian laser beam is higher at the center than at the edges, the beam experiences a higher refractive index at the center than at the periphery, provided $\chi^{(3)} > 0$. This spatial variation of the refractive index acts exactly like a graded index lens, shown in Figure 5. The medium bends the wavefronts toward the high intensity axis. This phenomenon, known as self-focusing, is governed by the effective nonlinear polarization [5]:

$$\mathbf{P}_{\text{NL}} = \frac{3}{4} \varepsilon_0 \chi^{(3)} |\mathcal{E}|^2 \mathbf{E} \tag{17}$$

The phase of a femtosecond laser pulse in a dielectric depends on the nonlinear refractive index, $\Phi(t) = n k_0 z - \omega_0 t = n_0 k_0 z + n_2 k_0 z I - \omega_0 t$. Differentiating with respect to time gives the instantaneous frequency shift due to self-phase modulation (SPM):

$$\frac{d\Phi}{dt} = -\omega_0 + n_2 z \frac{\omega_0}{c} \frac{dI(t)}{dt} \tag{18}$$

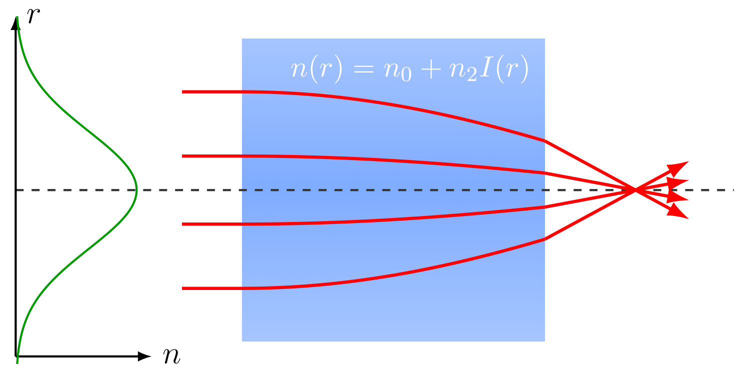


Figure 5: Kerr self-focusing as a graded index lens. The refractive index $n(r) = n_0 + n_2 I(r)$, plotted in green on the left, follows the transverse intensity profile of the beam, higher on axis where the beam is brightest (shaded blue gradient). Rays entering parallel to the axis are bent progressively more strongly as they approach the center, exactly as in a conventional graded index lens, and converge toward a nonlinear focus.

On the rising edge of the pulse, $dI/dt > 0$ and the instantaneous frequency is shifted toward the red, while on the falling edge, $dI/dt < 0$ and it is shifted toward the blue. This self-phase modulation alone, within the slowly varying envelope approximation, does not distort the temporal profile of the pulse. It only adds a frequency drift without changing the symmetric shape of $I(t)$.

Self-steepening is a distinct effect, which in the master propagation equation is intrinsically contained within the instantaneous Kerr term through the application of the operator $\hat{T}^2$. By defining $\hat{T} = 1 + \frac{i}{\omega_0}\frac{\partial}{\partial t}$, the expansion of $\hat{T}^2$ explicitly reveals the different physical components driving the nonlinear response:

$$
\begin{aligned}
i\hat{T}^2 \frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\,|u|^2u = \; & \underbrace{i\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\,|u|^2u}_{\text{Self-phase modulation (SPM)}} \\
& - \underbrace{\frac{2}{\omega_0}\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\frac{\partial}{\partial t}\left(|u|^2u\right)}_{\text{Self-steepening}} \\
& - \underbrace{\frac{i}{\omega_0^2}\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\frac{\partial^2}{\partial t^2}\left(|u|^2u\right)}_{\text{Higher order}}
\end{aligned}
$$

Using the equivalence between the envelope intensity and the electric field

$$
\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\,|u|^2 \equiv \frac{\omega_0 n_2}{c}|\mathcal{E}|^2
$$

This first-order temporal derivative evaluates exactly to the classic optical shock term $\frac{2n_2}{c}\frac{\partial}{\partial t}\left(|\mathcal{E}|^2\mathcal{E}\right)$. This shock originates from the fact that the group velocity itself depends on intensity through $n_2 I$ [14]. Because the peak of the pulse is more intense, its local velocity is smaller than that of the lower-intensity regions. As a result, the peak is slowed down relative to the overall group velocity, while the back of the pulse catches up with it. This dynamic compresses the energy at the rear, creating a sharp temporal gradient : the self-steepening effect [9]. This effect distorts the temporal profile in an intrinsically asymmetric way, even for a perfectly symmetric Gaussian input pulse. Because the intensity drops much more abruptly at the back of the pulse, the time derivative dI/dt in that region is strongly enhanced. Consequently, the SPM-induced blue-shift dominates the red-shift, leading to a strongly asymmetric spectrum with a pronounced blue extension—a hallmark of supercontinuum generation [14, 10]. In the spatio-temporal domain, this effect, coupled with dispersion, ultimately drives asymmetric pulse splitting [9]. Furthermore, in our numerical implementation, applying the full $\hat{T}^2$ operator directly in the Fourier domain (via the ω/ω_0 multiplier) avoids any Taylor truncation, ensuring that all dispersive orders of the self-steepening effect are exactly captured.

2.5 Plasma Defocusing

Several effects generate electrons in the conduction band. These include thermal excitation, tunnel ionization, multiphoton ionization, and free electrons resulting from crystal defects. All these free carriers respond to the incident electric field. The simplest way to model this is to treat the conduction electrons as a free electron gas subject to friction $\gamma = 1/\tau$. The equation of motion in a field $\vec{E}(t) = \vec{E}_0\, e^{-i\omega_0 t}$ reads:

$$m^* \frac{d^2}{dt^2}\vec{x} = -e\vec{E} - \frac{m^*}{\tau}\frac{d}{dt}\vec{x} \Longrightarrow \vec{x} = \frac{\tau e \vec{E}}{m^* \omega_0\,(\omega_0 \tau + i)} \tag{19}$$

The polarization of the free carriers $\vec{P}_{\text{free}} = -eN\vec{x}$ follows:

$$\vec{P}_{\text{free}} = -eN\vec{x} = -\frac{\omega_p^2 \tau}{\omega_0\,(\omega_0\tau + i)}\varepsilon_0 \vec{E} \tag{20}$$

where we introduce the plasma frequency $\omega_p^2 = \dfrac{Ne^2}{\varepsilon_0 m^*}$.

The valence electrons are not removed by the excitation, and their polarization $\vec{P}_{\text{bound}} = \varepsilon_0 \chi^{(1)} \vec{E}$ must be conserved. Both populations are driven by the same macroscopic field and respond linearly, so their contributions add up. Introducing the linear index $n_0^2 = 1 + \chi^{(1)}$ of the unexcited medium, the relative permittivity reads:

$$\varepsilon_r = \frac{D}{\varepsilon_0 E} = \frac{1}{\varepsilon_0 \vec{E}}\left(\varepsilon_0\vec{E} + \vec{P}_{\text{bound}} + \vec{P}_{\text{free}}\right) = n_0^2 - \frac{\omega_p^2 \tau}{\omega_0(\omega_0\tau + i)} \tag{21}$$

Multiplying numerator and denominator by the complex conjugate and separating the real and imaginary parts, we obtain:

$$\varepsilon_r = \varepsilon_1 + i\varepsilon_2 \tag{22}$$

$$\varepsilon_1 = n_0^2 - \frac{\omega_p^2 \tau^2}{1 + \omega_0^2 \tau^2}, \tag{23}$$

$$\varepsilon_2 = \frac{\omega_p^2 \tau}{\omega_0\left(1 + \omega_0^2\tau^2\right)} > 0, \tag{24}$$

The complex index follows from $\tilde{n} = n + i\kappa = \sqrt{\varepsilon_r}$, so $\varepsilon_1 = n^2 - \kappa^2$ and $\varepsilon_2 = 2n\kappa$.

When the number of oscillations before a collision is large, the real part of the refractive index simplifies to:

$$n = \sqrt{\varepsilon_1} = \left(n_0^2 - \frac{\omega_p^2\tau^2}{1+\omega_0^2\tau^2}\right)^{\frac{1}{2}} \approx n_0 \left(1 - \frac{\omega_p^2}{n_0^2\,\omega_0^2}\right)^{\frac{1}{2}} \tag{25}$$

The square root must be real, which requires $\dfrac{\omega_p^2}{n_0^2\,\omega_0^2} \leq 1$. Expanding the definition of the plasma frequency, we obtain:

$$\frac{Ne^2}{n_0^2\,\varepsilon_0 m^* \omega_0^2} \leq 1 \implies N \leq \frac{n_0^2\,\varepsilon_0 m^* \omega_0^2}{e^2} \tag{26}$$

Defining the critical density N_c as the density at which the plasma frequency equals the laser frequency, $\omega_p(N_c) = \omega_0$, we have:

$$N_c = \frac{\varepsilon_0 m^* \omega_0^2}{e^2} \implies \frac{\omega_p^2}{\omega_0^2} = \frac{N}{N_c} \tag{27}$$

The condition then becomes $N \leq n_0^2 N_c$, beyond which the wave is evanescent and the plasma becomes reflective. The index is then:

$$n = n_0\sqrt{1 - \frac{\omega_p^2}{n_0^2\,\omega_0^2}} = n_0\sqrt{1 - \frac{N}{n_0^2 N_c}} \tag{28}$$

For an underdense plasma, $N \ll n_0^2 N_c$, the real part of the refractive index can be expanded as:

$$n \simeq n_0 - \frac{N}{2\,n_0 N_c} \tag{29}$$

This expression and the definition of the critical density N_c match the Drude model as it is used to describe beam defocusing by the free electron plasma generated by the pulse [5].

2.6 Filamentation

A low-intensity Gaussian beam propagating through a transparent medium converges and focuses at a specific point before diverging. The entire process is symmetric about the focal point: energy density increases up to the focus and then decreases. The Rayleigh length, denoted as $z_R = \frac{\pi n_0 w_0^2}{\lambda_0}$, represents the characteristic propagation distance over which a Gaussian beam's radius expands by a factor of $\sqrt{2}$. At higher intensities, near the focusing point, the Kerr effect causes the beam to focus more rapidly while distorting the wavefront. This high intensity near the focus point can balance the diffractive effects given by Gaussian propagation laws. When these two competing processes balance each other out, the beam can maintain itself as a narrow, high-intensity channel over a distance far exceeding what standard focusing would allow. Whether the beam tends to diverge or converge is determined by the ratio of the incident power P_{in} to a critical power P_{cr}.

The critical power P_{cr} corresponds to the value of P_{in} at which self-focusing and diffraction exactly cancel each other out. Thus, in a model free of losses or external perturbations, and neglecting plasma effects, a beam injected with power $P_{in} = P_{cr}$ would propagate indefinitely with a constant radius. If $P_{in} < P_{cr}$, diffraction dominates, and the beam undergoes ordinary propagation. $P_{in} > P_{cr}$: self-focusing wins, the intensity increases, free carriers gets promoted within the conduction band and the plasma defocusing come into play, and can lead to cycles of focalisation defocalisation.

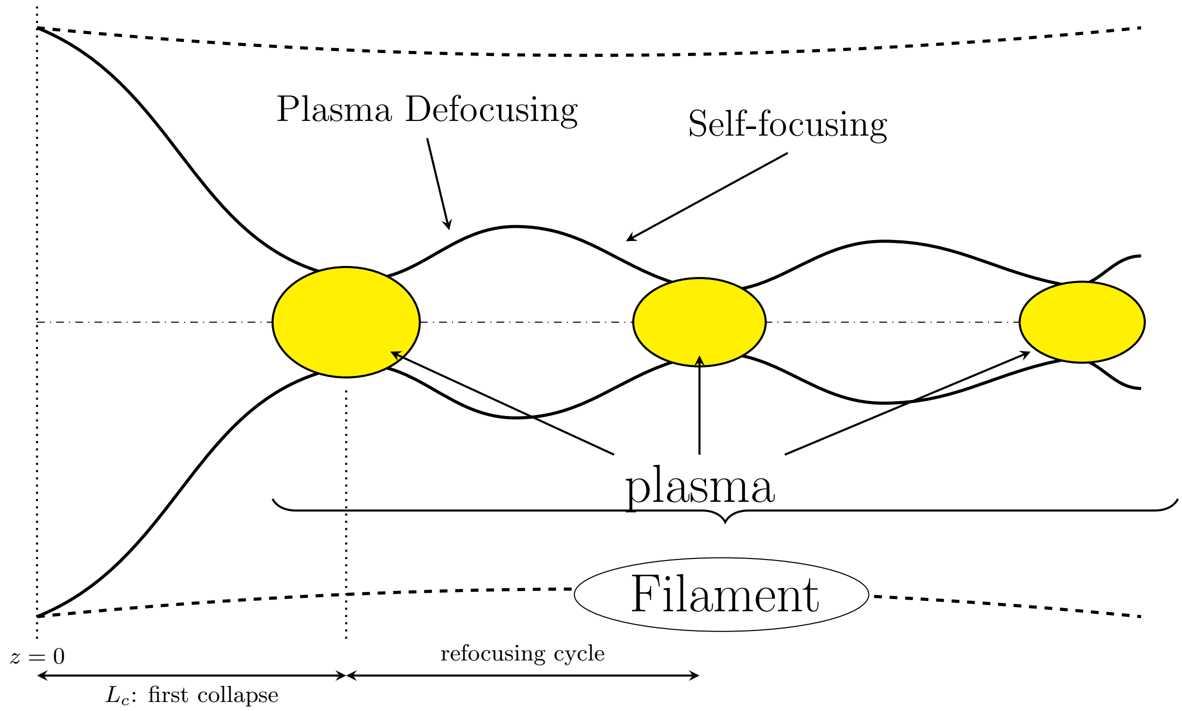


Figure 6: Schematic beam envelope undergoing repeated cycles of Kerr self-focusing and plasma defocusing, starting from an input beam launched at $z = 0$. The first collapse occurs at $z = L_c$, given by Eq. (**??**). Each collapse is arrested before the beam reaches a singular focus by the free electron plasma generated near the intensity peak (yellow), which defocuses the beam until the intensity drops enough for the plasma to recombine and self-focusing to take back over, refilled by the surrounding energy reservoir. This repeated breathing over an extended propagation distance is the filament. Figure **??** shows this same sequence of collapses in an actual simulation, rather than as a schematic.

2.7 Exciton Formation

When an electron is promoted from the valence band to the conduction band, for example by optical absorption, it leaves behind a hole of effective charge $+e$. Electrons in the conduction band can feel the potential created by the holes in the valence band. This electrostatic attraction can be modeled by a Coulomb potential, which allows us to solve the Schrödinger equation for a composite particle called an exciton, made of a conduction band electron and a valence band hole.

This quasiparticle has a reduced mass defined by $\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_h}$ and a quantized energy spectrum of the hydrogen-like type.

In the absence of Coulomb interaction, the minimum energy required to create a free electron-hole pair corresponds to the gap energy E_g. However, because of the attractive Coulomb potential between the electron and the hole, they can form a bound state. The accessible energy levels then sit below the conduction band. These states can be populated directly through the process described above, but also through nonradiative relaxation mechanisms such as the Auger effect or exciton-phonon interactions. As pointed out by Mao et al. [5], this formation process is very fast, often under 1 ps for wide bandgap materials.

The relative motion of this electron-hole pair can be modeled by a hydrogen-like Schrödinger equation. The stationary equation for the relative coordinate $\mathbf{r}$ reads:

$$\left[\frac{\hbar^2}{2\mu}\nabla^2 + \frac{e^2}{4\pi\epsilon_0\epsilon_r r}\right]\psi(\mathbf{r}) = E_b\psi(\mathbf{r}), \tag{30}$$

where $\epsilon = \epsilon_0 \epsilon_r$ is the dielectric permittivity of the medium and $E_b > 0$ is the exciton binding energy.

The relation between the gap energy E_g, the binding energy term, and the excitation energy E_n required for the n-th exciton state is given by:

$$E_g = E_n + \frac{R_\chi}{n^2}, \tag{31}$$

where R_χ is the effective exciton Rydberg energy:

$$R_\chi = \frac{\mu e^4}{2(4\pi\epsilon)^2\hbar^2} = R_H \left(\frac{\mu}{m_0}\right)\frac{1}{\epsilon_r^2}. \tag{32}$$

Here, $R_H \approx 13.6$ eV is the Rydberg constant of the hydrogen atom, m_0 is the free electron mass, and ϵ_r is the relative dielectric constant of the material.

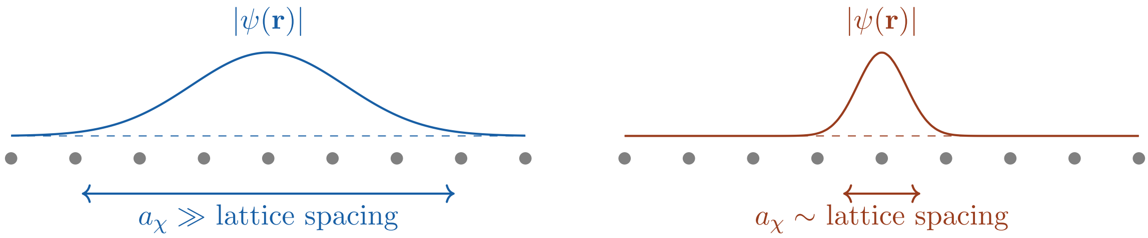
To evaluate the spatial extent of the exciton wavefunction, we define the exciton Bohr radius a_χ:

$$a_\chi = \frac{4\pi\epsilon\hbar^2}{\mu e^2} = a_0 \cdot \epsilon_r \left(\frac{m_0}{\mu}\right), \tag{33}$$

where $a_0 \approx 0.53$ Å is the atomic Bohr radius.

In wide bandgap materials, electrons are strongly bound to their atoms, which means the relative dielectric permittivity is low for SiO_2, where the relative dielectric constant is typically of order 2 to 4. As a result, the material provides weak dielectric screening of the Coulomb attraction, maintaining a deep potential well.

A low permittivity directly leads to a reduction of the exciton Bohr radius, down to a few angstroms. When a_χ becomes comparable to the interatomic spacing of the lattice, the exciton localization no longer extends over several lattice atoms but shifts to a strongly bound regime, illustrated in Figure 7. The binding energy also increases, preventing thermal dissociation and favoring ultrafast energy relaxation. During this process, the excess electrostatic potential energy of the unscreened exciton is abruptly discharged into local lattice vibrations (phonons) through massive exciton-phonon coupling within a few picoseconds. This induces a local lattice distortion and leads to the formation of a self-trapped exciton (STE).



(a) Weakly bound exciton, large permittivity ϵ_r: the wavefunction envelope extends over many lattice sites, close to a free electron-hole pair.

(b) Strongly bound exciton, low permittivity: the wavefunction collapses onto a single lattice site, the regime relevant for SiO_2.

Figure 7: Spatial extent of the exciton wavefunction relative to the lattice spacing. The grey dots represent lattice atoms along one direction, the colored curve the envelope $|\psi(\mathbf{r})|$ of the relative electron-hole wavefunction. A smaller ϵ_r shrinks a_χ from Eq. (33) until it reaches the strongly bound regime on the right.

2.8 Self-Trapped Excitons

Thermal fluctuations induce slight lattice vibrations that act as a trigger, creating a transient local deformation. The strongly localized exciton is captured by this distortion. Through massive exciton-phonon coupling, it abruptly discharges its potential energy into the lattice, considerably deepening the local potential well, shown schematically in Figure **??**.

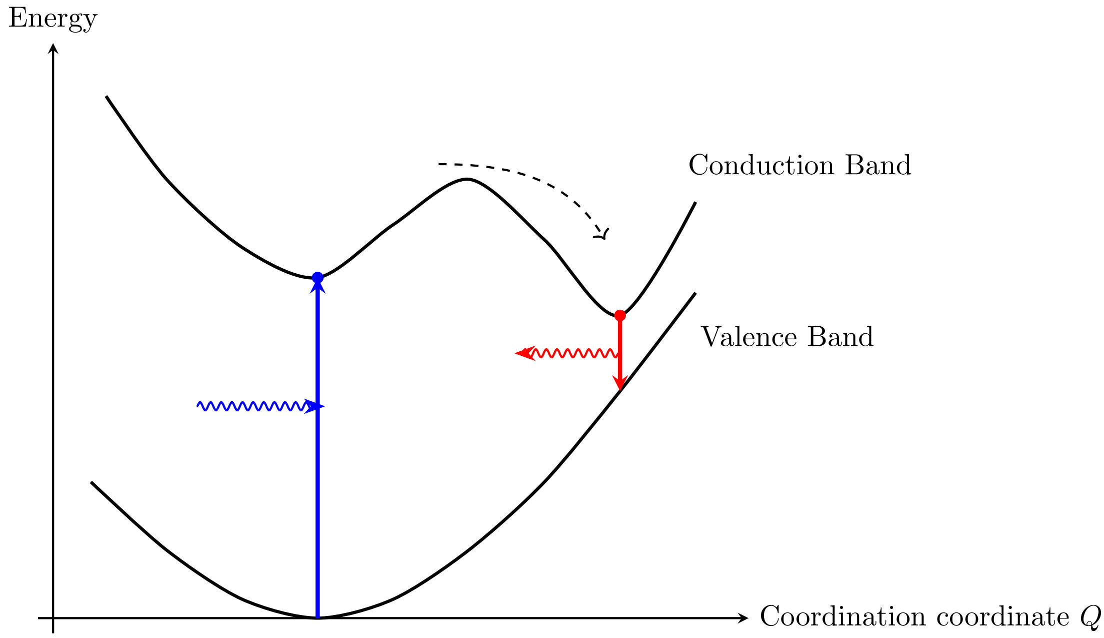
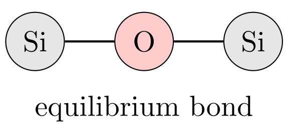
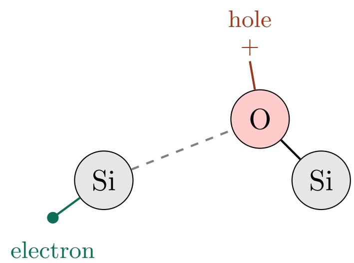


Figure 8: Configuration-coordinate diagram illustrating the formation of a self-trapped exciton (STE). After photoexcitation, a free exciton relaxes non-radiatively through lattice distortion into the STE configuration before radiative recombination.

This intense localized energy weakens and breaks the Si–O–Si covalent bond, displacing the oxygen atom from its equilibrium position. This structural rupture generates two dangling bonds, each holding an unpaired electron. In this strongly distorted configuration, the exciton self-traps. The hole localizes on the oxygen dangling bond, while the electron remains bound to the silicon dangling bond [5], as sketched in Figure 9.



(a) Before: intact Si–O–Si bridge, oxygen at its equilibrium position.



(b) After: the Si–O bond breaks, the oxygen relaxes off site, leaving a silicon dangling bond carrying the electron and an oxygen dangling bond carrying the hole.

Figure 9: Schematic bond-breaking mechanism of self-trapping in SiO_2, simplified to a linear Si–O–Si fragment for clarity (the real network is tetrahedrally coordinated). The two dangling bonds left behind are the microscopic identity of the self-trapped exciton.

This mechanism is responsible for the formation of Self-Trapped Excitons in silica. The excitation energy is rapidly converted into local lattice distortion and permanent structural modifications of the material [5].

3 Propagation Equation and Population Rate Equations

The electric field is written as a carrier wave modulated by a slowly varying envelope $\mathcal{E}(r,t,z)$, and the wave equation is reduced under the slowly varying envelope approximation. The envelope is normalised so that its squared modulus is the local intensity,

$$|\mathcal{E}|^2 = I = \tfrac{1}{2} n_0 c \varepsilon_0 |u|^2, \tag{34}$$

where u is the field amplitude in volts per metre that the code actually stores. Choosing $|\mathcal{E}|^2$ to be an intensity is what makes every coefficient below a plain inverse length, which is the form used by Couairon et al. [8]. With that convention the propagation equation solved by the code reads:

$$\begin{aligned}
\hat{U}\,\frac{\partial \mathcal{E}}{\partial z} = {} & \underbrace{\frac{i}{2k_0}\nabla_\perp^2 \mathcal{E}}_{\text{diffraction}} + \underbrace{i\,\hat{D}\,\hat{U}\,\mathcal{E}}_{\substack{\text{dispersion: exact Sellmeier, all orders}\\ \text{leading term } -\frac{i}{2}k''_\omega\,\partial_t^2\mathcal{E}}} \\
& + \underbrace{i\,k_{vac}\,n_2\,\hat{T}^2\Big[(1-f_R)\,|\mathcal{E}|^2 + f_R\big(R * |\mathcal{E}|^2\big)\Big]\mathcal{E}}_{\text{Kerr effect, instantaneous and delayed Raman, with self-steepening}} \\
& - \underbrace{\frac{\hat{T}}{2}\,\frac{U_i\,W_{PI}(I)}{I}\left(1-\frac{N}{N_{at}}\right)\mathcal{E}}_{\text{photoionization energy loss}} - \underbrace{\frac{\sigma_\omega}{2}\big(1+i\omega_0\tau_c\big)N\,\mathcal{E}}_{\text{plasma absorption and defocusing}} \\
& + \underbrace{i\,k_{vac}\,\frac{1}{2n_0\rho_c}\,\frac{\omega_0^2}{\omega_{tr}^2-\omega_0^2}\,N_{STE}\,\mathcal{E}}_{\text{bound STE polarizability, see Section 11.4}}
\end{aligned} \tag{35}$$

The two wavenumbers appearing here are not the same quantity, and keeping them apart is the whole point of writing the equation this way:

$$k_0 = \frac{n_0\omega_0}{c} \quad \text{(in the medium)}, \qquad k_{vac} = \frac{\omega_0}{c} = \frac{k_0}{n_0} \quad \text{(in vacuum)}. \tag{36}$$

Diffraction and the operator $\hat{U}$ use the wavenumber in the medium, because they describe how a wave already inside the glass spreads and how its group delay varies. The two index changing terms use the vacuum wavenumber, because a change of refractive index Δn over a length L produces a phase $\Delta\varphi = (\omega_0/c)\,\Delta n\,L$, with no extra factor of n_0. Writing both as k_0 would make the Kerr term too large by $n_0 = 1.45$.

The remaining symbols are

$$n_2 : \Delta n = n_2 I, \qquad \sigma_\omega : \text{Drude cross section, Section 7}, \qquad \rho_c = \frac{\varepsilon_0 m_e \omega_0^2}{e^2}, \tag{37}$$

The Keldysh rate is written as a function of the intensity, $W_{PI}(I)$, because that is how it is tabulated and how the solver calls it. The rate is of course driven by the field amplitude, through the Keldysh parameter $\gamma = \omega_0\sqrt{m^* U_i}/(eE)$, but that conversion happens inside `KeldyshSiO2.rate`, which takes an intensity, recovers $E = \sqrt{2I/n_0 c\varepsilon_0}$ and builds γ from it. Writing $W_{PI}(|u|)$ would name a different function of a different argument.

The operators, in the retarded frame $t \to t - z/v_g$ with $\Omega = \omega - \omega_0$, are

$$\hat{T} = 1 + \frac{i}{\omega_0}\frac{\partial}{\partial t} \;\longleftrightarrow\; 1 + \frac{\Omega}{\omega_0}, \qquad \hat{U} = 1 + \frac{ik'_\omega}{k_0}\frac{\partial}{\partial t} \;\longleftrightarrow\; 1 + \frac{k'_\omega \Omega}{k_0}, \tag{38}$$

$$\hat{D} \longleftrightarrow k(\omega) - k_0 - k'_\omega \Omega, \tag{39}$$

where $k(\omega)$ comes from the Sellmeier relation and is not truncated at any order.

Every coefficient in Eq. (35) is an inverse length, which makes the equation easy to check. In the Kerr term $n_2 I$ is a dimensionless index change and k_{vac} carries the inverse metre. In the photoionization term $U_i W_{PI}$ is an absorbed power per unit volume, in watts per cubic metre, and dividing by the intensity in watts per square metre leaves an inverse metre. In the plasma term σ_ω is an area and N a volume density, so their product is again an inverse metre. The factor of one half on the last two terms is not cosmetic. Those two channels remove energy, and the equation propagates an amplitude rather than an intensity, so a coefficient α acting on $\mathcal{E}$ attenuates I at the rate 2α. The Kerr and STE terms carry no such factor because they are pure phases and remove nothing.

The relation to the susceptibility form is $n_2 = 3\chi^{(3)}/(4n_0^2 c\varepsilon_0)$, already given in Section 6. The code stores $\chi^{(3)}$ internally and builds the prefactor $3\omega_0^2\chi^{(3)}/(8k_0c^2)$, which multiplies $|u|^2$ rather than $|\mathcal{E}|^2$. Substituting Eq. (34) turns it into $k_{vac}n_2$ exactly, and the script `check_prefactors.py` verifies the identity numerically to twelve digits against the constants the solver is given.

This is the form used by Couairon et al. [8]. Their Eq. (2) is introduced with exactly that wording, namely that the equation accounts for space-time focusing through the operator $\hat{U}$ sitting in front of $\partial/\partial z$. The reason to prefer this form over the algebraically equivalent one obtained by dividing by $\hat{U}$ is that it contains no inverse operator, so there is no ambiguity about which terms $\hat{U}$ acts on. If one does divide by $\hat{U}$, it has to be applied to every term on the right hand side, which gives

$$\frac{\partial \mathcal{E}}{\partial z} = \frac{i}{2k_0}\hat{U}^{-1}\nabla_\perp^2 \mathcal{E} + i\hat{D}\mathcal{E} + \hat{U}^{-1}\Big\{\text{Kerr} + \text{Raman} + \text{ionization} + \text{plasma} + \text{STE}\Big\}, \tag{40}$$

where $\hat{U}$ cancels on the dispersion term alone. Since $\hat{U} \simeq \hat{T}$ to within one percent for fused silica at 800 nm (with $n_g/n_0 = 1.0095$), the Kerr operator in that form becomes $\hat{U}^{-1}\hat{T}^2 \simeq \hat{T}$ and the ionization operator becomes $\hat{U}^{-1}\hat{T} \simeq 1$, which is how the single $\hat{T}$ expressions found in review papers arise. Applying $\hat{U}^{-1}$ to the Laplacian only, while leaving $\hat{T}^2$ on the Kerr term, is not one of these two forms and leaves every nonlinear term with one power of $\hat{T}$ too many.

Two smaller points about notation are worth stating clearly. First, the depletion factor is the dimensionless $1 - N/N_{at}$, not $N_{at} - N$. The Keldysh rate W_{PI} returned by the lookup table is already a volumetric rate in $\text{cm}^{-3}\text{s}^{-1}$, so multiplying it by a density would leave the term with an extra cm^{-3}. The other convention, in which W_{PI} is a per atom rate in s^{-1} multiplied by $N_{at} - N$, is equally valid but is not what the code computes. Second, the group delay $k'_\omega\Omega$ does not appear as a separate term because it has been absorbed into $\hat{D}$. The code works in the retarded frame and subtracts it inside `delta_k`, so writing it again outside would count it twice.

Equation (35) cannot be stepped forward as written, because it gives $\hat{U}\partial_z\mathcal{E}$ rather than $\partial_z\mathcal{E}$. The code therefore integrates the divided form (40). The linear half step in `LinearOperator.half_linear` supplies the first two terms, applying `inv_U_op` to the Laplacian and leaving `delta_k` bare. The nonlinear step in `NonlinearOperator.split` assembles the whole brace in frequency space and multiplies it by $\hat{U}^{-1}$ in one operation, so that the Kerr, Raman, ionization, plasma and STE terms all receive it. Because $\hat{U}^{-1}$ here multiplies source terms directly rather than sitting inside a phase factor $e^{i\varphi}$, it is floored and masked before use. The operator $\hat{U}$ vanishes at $\Omega = -k'^{-1}_\omega k_0 \simeq -0.99\,\omega_0$, that is at essentially zero absolute frequency. The floor only acts beyond $\lambda = 7.4$ μm, outside the Sellmeier window where the spectral mask has already suppressed the field, so within the physical band $\hat{U}^{-1}$ stays between 0.22 and 6.6 and is not distorted.

3.1 Where the Self-Steepening Operator Comes From, and Why It Is Squared on Some Terms

Section 6 introduced self-steepening through the time domain shock term $(2n_2/c)\,\partial(|\mathcal{E}|^2\mathcal{E})/\partial t$. The propagation equation above instead writes it as a frequency domain multiplication by $T = 1+\Omega/\omega_0$, or by T^2 on some terms. These are the same physics in two different languages. The reason some terms get T, some get T^2, and the plasma term gets no such operator at all, comes directly from how many time derivatives that term carries before the slowly varying envelope approximation is applied.

Every source term on the right hand side of the underlying wave equation comes from a current or a polarization evaluated at the true frequency $\omega = \omega_0 + \Omega$ of each spectral component, not at the fixed carrier frequency ω_0. Writing $\omega = \omega_0(1 + \Omega/\omega_0) = \omega_0 T$, any factor of ω appearing in a source term becomes a factor of $\omega_0 T$ once it is pulled out and referenced to the carrier frequency, in the same way that $k(\omega)$ was expanded around k_0 for the dispersion term. The Kerr polarization enters the wave equation through $\partial^2 P_{NL}/\partial t^2$, a second time derivative, which in the frequency domain is a factor of $-\omega^2 = -\omega_0^2 T^2$. The photoionization current, like the plasma current, enters through a single time derivative $\partial J/\partial t$, a factor of $-i\omega = -i\omega_0 T$. That is the whole reason the Kerr and Raman terms carry T^2 while the photoionization loss term carries only T. Both are the same operator $T = 1 + \Omega/\omega_0$, applied once for every power of ω that the term's own derivation contributes.

The plasma current J_{plasma} also comes from a single time derivative, so by the same counting it should carry T^1 as well. It appears without any T in the equation above because the Drude derivation of Section 7 already keeps the full frequency dependence of $\sigma(\omega)$ and $(1 + i\omega\tau_c)$ exactly, instead of expanding around ω_0 first. The correction is therefore already built into σ_ω itself and does not need to be reintroduced as a separate multiplicative operator.

Physically, T being larger than one on the blue side of the spectrum ($\Omega > 0$) and smaller than one on the red side is the same statement as the shock term of Section 6 pulling energy toward the trailing edge of the pulse. Whichever nonlinear term this operator multiplies, it systematically favours the blue shifted, trailing part of the spectrum over the red shifted, leading part, which is the frequency domain signature of the group velocity depending on intensity. Squaring the operator on the Kerr term simply means this asymmetry is applied twice, once for each power of ω that the second time derivative of P_{NL} contributes. This is why the Kerr induced spectral asymmetry is normally the dominant contribution to self-steepening compared with the single power of T on photoionization loss.

Each underbraced term maps onto a specific piece of the code. Diffraction and dispersion are applied together by `LinearOperator.half_linear`, in the quasi discrete Hankel basis described in Section 12. The space-time focusing factor enters this same step as `inv_U_op` $= 1/\hat{U}(\Omega)$ multiplying the transverse Laplacian, and reduces to the identity when `enable_space_time_focusing` is off. Rather than truncating the Taylor expansion of $k(\omega)$ at second order as was done analytically in Section 6, the code evaluates $k(\omega)$ directly from the Sellmeier relation for every frequency component and subtracts the linear term $k_0 + k'_\omega\Omega$, so the operator `delta_k` already contains dispersion to all orders and not just group velocity dispersion.

The Kerr, self-steepening and Raman terms are computed together, inside the function `split` of `NonlinearOperator`. The instantaneous part $(1 - f_R)|\mathcal{E}|^2$ and the delayed Raman part $f_R\,(R * |\mathcal{E}|^2)$, obtained by convolving the intensity with the retarded Raman response function $R(t)$, are summed into a single quantity `kerr_I` before being multiplied by the field, which is why they appear here as two lines of one mechanism rather than as two independent effects. The self-steepening operator is applied in the frequency domain as `T_op` $= 1 + \Omega/\omega_0$, following the same operator introduced by Mao et al. [5] for free carrier absorption and used identically here for the Kerr response.

The photoionization loss term removes energy from the field at exactly the rate needed to promote electrons to the conduction band at the Keldysh rate W_{PI}, throttled by the fraction $1 - N/N_{at}$ of ionizable sites still neutral. In the code this factor is `depl_field = clip(1 - rho/rho_max, 0, 1)`, where N_{at} is `rho_max` and N is the current free electron density. The plasma absorption and the plasma induced defocusing are written as one term above because the code computes them together, as the real and imaginary parts of the same Drude response derived in Section 7, with σ_ω the inverse Bremsstrahlung cross section introduced in Section 5.

Six boolean flags, one per mechanism, let each term be turned off on its own. They are called `enable_kerr_instantaneous`, `enable_kerr_raman`, `enable_self_steepening`, `enable_photoionization_loss`, `enable_plasma_defocusing` and `enable_plasma_absorption`. The two Kerr flags gate the instantaneous and Raman contributions to `kerr_I` separately rather than gating the shared prefactor, so switching one off keeps the correct $(1 - f_R)$ and f_R weights on the other. Turning off `enable_photoionization_loss` removes the loss from the field equation only. Carrier generation keeps running off the same Keldysh rate, so the plasma still forms and the beam still defocuses, which separates the energy cost of ionization from its optical effect. This is what the ablation study in the accompanying notebook uses to test what each term actually does to the beam, instead of only reading it off the equation.

The solver also keeps a spectral mask, `spec_mask`, multiplying `T_op`. The reason is that $T = 1+\Omega/\omega_0$ becomes negative for absolute frequencies below zero, and this sign flip was found to be numerically unstable, amplified by up to a factor of nine per step once the operator is squared. The mask is a smooth tanh window that suppresses the field outside the wavelength range where the Sellmeier fit is valid, $\lambda \in [0.18, 5]$ μm, and is controlled by `enable_spectral_filter`, on by default. It has no effect on the physical, non negative frequency part of the spectrum.

3.2 Carrier Density Rate Equations

The code integrates two coupled populations, the free electron density N and the self-trapped exciton density N_{STE}, following the two population model of Mao et al. [5]:

$$\begin{aligned}\frac{dN}{dt} = {} & \underbrace{W_{PI}(I)\left(1 - \frac{N}{N_{at}}\right)}_{\text{direct photoionization, depletes available sites}} \\ & + \underbrace{\Big[W_{STE}(I) + \beta_s I\, N\Big]\frac{N_{STE}}{N_{at}}}_{\text{re-ionization of trapped excitons}} \\ & + \underbrace{\beta_g\, I\, N\left(1 - \frac{N}{N_{at}}\right)}_{\text{avalanche ionization}} - \underbrace{\frac{N}{\tau_r}}_{\text{trapping into the STE channel}} \end{aligned} \tag{41}$$

$$\frac{dN_{STE}}{dt} = \underbrace{\frac{N}{\tau_r}}_{\text{trapping}} - \underbrace{\Big[W_{STE}(I) + \beta_s I\, N\Big]\frac{N_{STE}}{N_{at}}}_{\text{re-ionization}} - \underbrace{\frac{N_{STE}}{\tau_{STE}}}_{\text{decay to the ground state}} \tag{42}$$

Here $I = |\mathcal{E}|^2$ is the local intensity, W_{STE} is the same Keldysh formula evaluated at the smaller gap U_s of the trapped state instead of U_i, and $\beta_g = \sigma_\omega/U_i$ is the avalanche coefficient with β_s its counterpart for the STE channel. In the CUDA kernel these two coefficients appear as `beta_g` and `beta_s`, the trapping rate as `inv_tau_r`, and the decay rate as `inv_tau_ste`.

The two populations exchange carriers in both directions. Trapping moves electrons out of the conduction band into the STE state at the rate N/τ_r, and that same quantity reappears as a gain term in the second equation, so nothing is lost in the transfer. Laser driven re-ionization sends them back the other way, and again the term appears with opposite signs in the two equations. Only the last term of the second equation, $-N_{STE}/\tau_{STE}$, genuinely removes carriers from the pair of populations, since it returns them to the ground state. The kernel also rescales both populations if their sum ever exceeds N_{at}, which keeps the model from producing more carriers than there are sites.

Note the correspondence with Section 5. There, the same pair of equations was written with σ/E_g in place of β_g and τ_x in place of τ_r. These are the same quantities, and the notation has only been aligned here with the variable names used in the code.

The two loss terms of the second equation are easy to read as the same thing written twice, so it is worth separating them. The trapping time τ_r, about 150 fs in fused silica, is how fast a conduction band electron falls into a self-trapped state. The relaxation time τ_{STE}, about 1 ps, is how long that trapped state then survives before decaying non radiatively to the ground state. The first controls how quickly the free electron population drains, the second controls how long the trapped population, and the index change it produces through Section 11.4, persists after the pump pulse has left. The second time is set by `Config.tau_ste` and defaults to `None`, which disables that channel entirely. Since the probe measurements of Section 14 extend to picosecond delays, this term matters for the late part of the phase curve even though it barely affects the pulse itself.

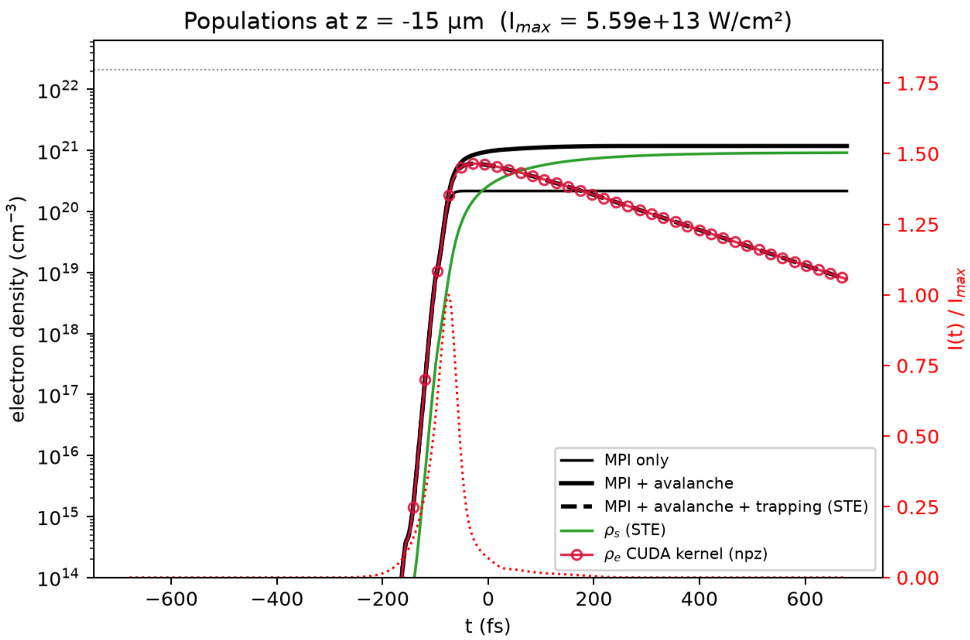


Figure 10: On-axis electron populations at the nonlinear focus for the 800 nm, 1.1 μJ case, reintegrating the 0D rate equations above from the saved intensity trace $I(t)$ under three models (MPI only, then with avalanche, then with avalanche and trapping) and superposing the $N(t)$ that the CUDA kernel actually computed on-axis (open circles). The full model coincides with the CUDA output by construction, which is what this overlay is meant to check, and the agreement confirms that the exponential time stepping of Section 12 reproduces the same rate equations solved here term by term. Trapping into N_{STE} (green) becomes the dominant channel within a few hundred femtoseconds of the pulse peak, draining N (dashed black) down from its avalanche augmented maximum.

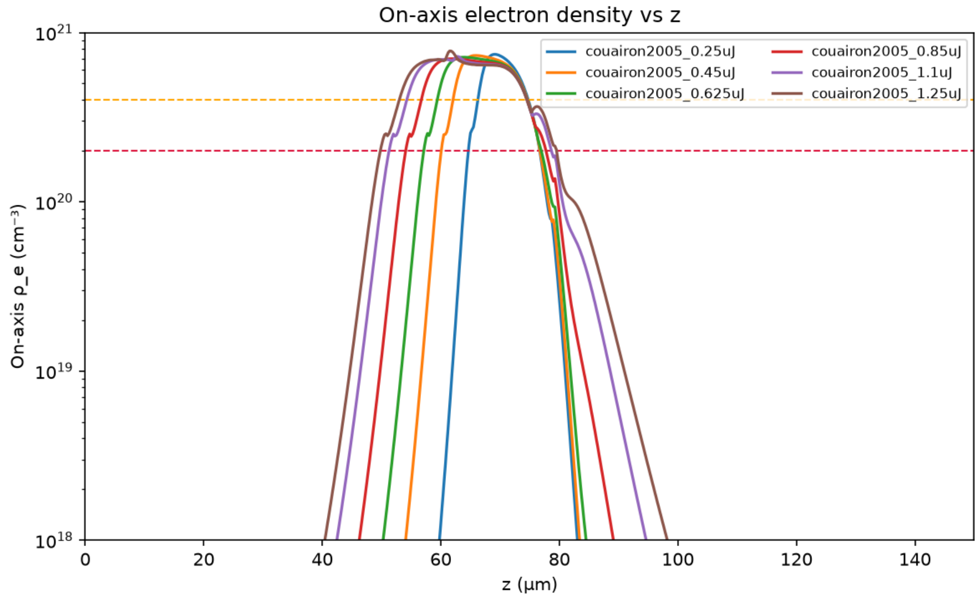


Figure 11: On-axis electron density along z for six input energies, reproducing Fig. 9 of [8]. The dashed lines mark 1 and 2×10^{20} cm^{-3} as reference levels. Every above-threshold curve saturates near a common density once the local intensity reaches I_{clamp}, which is the electron density signature of the clamping balance discussed in Section 8.

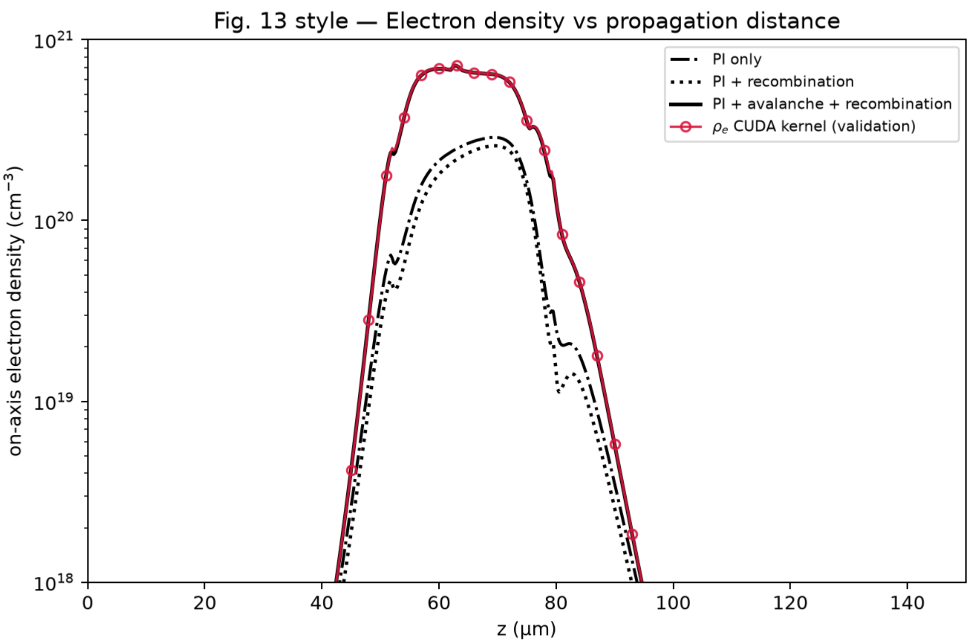


Figure 12: On-axis electron density along z at fixed input energy (1.1 μJ), comparing photoionization only, photoionization with recombination, and the full model with avalanche, against the CUDA kernel output (open circles), reproducing Fig. 13 of [8]. The full model (thick solid) tracks the CUDA output across the four decades shown, while the two reduced models fall visibly below it before the nonlinear focus, which isolates the contribution of the avalanche term $\beta_g IN$ to the total carrier yield.

3.3 Photoionization Energy Loss

For completeness we derive the photoionization loss term used above, since it does not appear explicitly in the earlier sections. Each ionization event removes one photoelectron from the neutral population and costs at least the gap energy U_i, which is taken directly from the field. The absorbed power per unit volume is

$$\mathcal{P}_{ion} = U_i \left(1 - \frac{N}{N_{at}}\right) W_{PI}(I) \tag{43}$$

The equation propagates an amplitude, not an intensity, so a coefficient α_{ion} acting on $\mathcal{E}$ attenuates I at the rate $2\alpha_{ion}$. Setting $2\alpha_{ion} I$ equal to the absorbed power density and solving gives the coefficient used above,

$$\alpha_{ion} = \frac{U_i\, W_{PI}(I)}{2I} \left(1 - \frac{N}{N_{at}}\right) \tag{44}$$

which is the `photo` variable computed inside `NonlinearOperator.split`. In the code the same expression appears as $U_i W_{PI}/(n_0 c\varepsilon_0 |u|^2)$, with no explicit one half, because $n_0 c\varepsilon_0 |u|^2$ is already $2I$ by Eq. (34). The factor is easy to lose, and the check that it is right is in `Integrator._record`, which weights the local energy loss by 2α when it accumulates the energy budget.

3.4 Bound Polarizability of Self-Trapped Excitons

A self-trapped exciton is not a conduction band carrier. It is a pair of dangling bonds, $\mathrm{Si}^\bullet$ and $\mathrm{O}^\bullet$, left behind after the Si-O-Si bond breaks, as described in Section 10, and its unpaired orbitals sit inside the gap rather than in the conduction band. It therefore does not drift in the field and does not contribute to the free carrier Drude response of Section 7. What it does instead is polarize as a bound oscillator with a resonance at the energy of its first excited level, following Mao et al. [5]:

$$\Delta n_{STE} = \frac{N_{STE}\, e^2}{2 n_0 m_e \varepsilon_0} \frac{1}{\omega_{tr}^2 - \omega^2} \tag{45}$$

This contribution enters the field equation as the fifth line of Eq. (35), controlled by `enable_ste_index`. It is implemented as a pure phase term with no associated loss, because the pump photon energy at 1.55 eV is far from the STE resonance, so there is no one photon absorption to account for on the pump itself. Since the pump lies below the resonance, $\omega_0 < \omega_{tr}$, the term is positive and adds to the Kerr self-focusing rather than opposing it. This is also the mechanism identified as the permanent, fracture free index change that Couairon et al. classify as type I damage.

The resonance energy needs care. Mao et al. give 5.2 eV as the peak of the STE transient absorption band, which is not the resonance frequency to use in the index formula above. For that they give the first excited level at $\omega_{tr} \approx 4.2$ eV, and this is the value stored in `E_tr_eV`. Using 5.2 eV instead changes the resonance denominator at a 490 nm probe wavelength by a factor of about 1.84, so the distinction is not a detail.

For the probe analysis of Section 14 this single band picture is replaced by the more complete dielectric function of Martin et al. [6], which uses two STE bands with their own oscillator strengths and widths rather than one, and which also accounts for the valence electrons removed from the ground state by ionization. That model is described where it is used, in Section 14.

4 Numerical Method

The propagation equation of Section 11 mixes a transverse diffraction term, a temporal dispersion term, and a nonlinear term that depends on the local field at every point in space and time. No single numerical scheme handles all three cheaply. The code therefore treats each mechanism with the tool best suited to it, and recombines them at every propagation step through operator splitting. This section explains the three pieces in turn, namely the radial basis used for diffraction, the time stepping scheme that combines them, and the population equations that have to be solved alongside the field.

4.1 Quasi Discrete Hankel Transform for the Radial Coordinate

The beam has cylindrical symmetry, so the transverse Laplacian $\nabla_\perp^2$ acting on a field that depends only on r reduces to $\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)$. The natural basis for this operator is not the Fourier basis but the Hankel transform of order zero, because a diffracting wave with cylindrical symmetry is a superposition of Bessel functions $J_0(\rho r)$ rather than complex exponentials. In this basis diffraction becomes a multiplication by a phase factor, in the same way that dispersion becomes a multiplication in the temporal Fourier domain.

The continuous Hankel transform cannot be evaluated on a computer, so the code uses the quasi discrete Hankel transform of Guizar-Sicairos and Gutierrez-Vega. The idea is to sample the field not on a uniform radial grid, but at the zeros $j_1, j_2, \ldots, j_N$ of the Bessel function J_0. On this particular grid the forward and inverse Hankel transforms both collapse to the same linear transformation, represented by a single symmetric matrix Y:

$$Y_{mn} = \frac{2}{j_N}\,\frac{J_0(j_m j_n/j_N)}{J_1(j_m)^2} \tag{46}$$

Multiplying a field sampled at the radii $r_m = j_m R/j_N$ by Y gives its Hankel transform sampled at the spatial frequencies $\rho_m = j_m/R$, and multiplying by Y a second time returns the original field, since Y is its own inverse. This is what `build_grids` constructs and what `LinearOperator.half_linear` uses, applying Y before and after the diffraction phase.

One practical consequence of this grid is that the radial samples are not evenly spaced. They are densest near the axis and spread out toward the edge of the box, which suits a beam whose interesting structure sits near $r = 0$. It also means that anything reading the radial profile afterwards has to use the same non uniform grid, which is why the Abel matrix of Section 14 is built from shell integrals over these specific radii rather than from a uniform trapezoidal sum.

4.2 Strang Splitting of the Propagation Step

The full propagation equation can be written schematically as $\partial_z u = (\mathcal{L} + \mathcal{N})u$, where $\mathcal{L}$ groups the linear operators, diffraction and dispersion, and $\mathcal{N}$ groups the nonlinear terms. Solving $\mathcal{L}$ alone and $\mathcal{N}$ alone are both cheap, since diffraction and dispersion are diagonal in the Hankel and Fourier bases and the nonlinear term is diagonal in real space. Solving $\mathcal{L}+\mathcal{N}$ together is not.

Strang splitting exploits this by approximating the exact evolution operator with a symmetric product:

$$e^{(\mathcal{L}+\mathcal{N})dz} \approx e^{\mathcal{L}\,dz/2}\, e^{\mathcal{N}\,dz}\, e^{\mathcal{L}\,dz/2} \tag{47}$$

The operators do not commute, so this product differs from the exact evolution, but the error is third order in dz per step instead of second order for the naive product $e^{\mathcal{L}dz}e^{\mathcal{N}dz}$.

The linear half step applies diffraction and dispersion together in a single operation rather than one after the other. This is exact and not an extra approximation, because once the field is expressed in the combined basis of Hankel radial modes and temporal frequencies Ω, both operators are diagonal and therefore commute. Concretely, `half_linear` multiplies by the QDHT matrix, takes the FFT along time, multiplies by the single combined phase

$$\exp\left[i\left(\hat{D}(\Omega)-\frac{\rho^2}{2k_0}\hat{U}^{-1}(\Omega)\right)\frac{dz}{2}\right], \tag{48}$$

applies the spectral mask, then transforms back. Doing it this way costs one FFT and inverse FFT pair per call instead of the two pairs a separate frequency resolved diffraction step would need on top of the dispersion step.

Inside the nonlinear part, the code separates two things that behave differently. The absorption coefficient $\alpha = \alpha_{ion} + \sigma_\omega N/2$, coming from photoionization loss and plasma absorption, only reduces the amplitude of u without exchanging energy between frequency components, so it is applied exactly as a multiplicative factor $u \to u\, e^{-\alpha\, dz/2}$ on each side of the rest of the nonlinear step. The remaining part, that is the Kerr effect, self-steepening, the dispersive part of the plasma response and the STE polarizability, is a genuine differential equation in z and is integrated with a fourth order Runge-Kutta scheme over the substep. Because the exponential factor already applies the zeroth order of the two dissipative terms, `split` adds αu back into the right hand side it returns, so that the two treatments do not double count the same loss.

The full sequence executed by `Integrator.step` for one propagation step is therefore a half linear step, a half absorption step, the four Runge-Kutta stages, another half absorption step, and a final half linear step. A radial mask is applied at the end to damp anything reaching the edge of the box.

4.3 Population Rate Equations on the GPU

The rate equations of Section 11.2 are solved separately from the field envelope, by marching along the temporal axis at fixed radius, once before every propagation step. The equations for N and N_{STE} are stiff, since the photoionization and avalanche rates can change by orders of magnitude within a single time sample, so a plain explicit Euler step would need an impractically small Δt. The code avoids this by treating the coupled pair as a locally linear system over each time interval $[t_j, t_{j+1}]$,

$$\frac{dx}{dt} = S + Lx \tag{49}$$

with the source S and the rate L evaluated from the trapezoidal average of the intensity dependent rates at t_j and t_{j+1}. This linear equation has a closed form solution,

$$x(t_j + \Delta t) = x(t_j)\, e^{L\Delta t} + S\,\Delta t\, \varphi_1(L\Delta t), \qquad \varphi_1(z) = \frac{e^z - 1}{z} \tag{50}$$

which is what the function `exact_exp_step` in the CUDA kernel implements, with a Taylor expansion of φ_1 when $L\Delta t$ is close to zero to avoid dividing zero by zero. This is an exact integrator for the frozen coefficient problem, so it does not accumulate truncation error from the time stepping itself. The only error comes from assuming S and L are constant over each short interval, which is a much better approximation than freezing the whole nonlinear right hand side of a generic ordinary differential equation.

4.4 The Main Propagation Loop

Figure 13 puts the three preceding subsections together into the loop that `Integrator.propagate` runs once per z step.

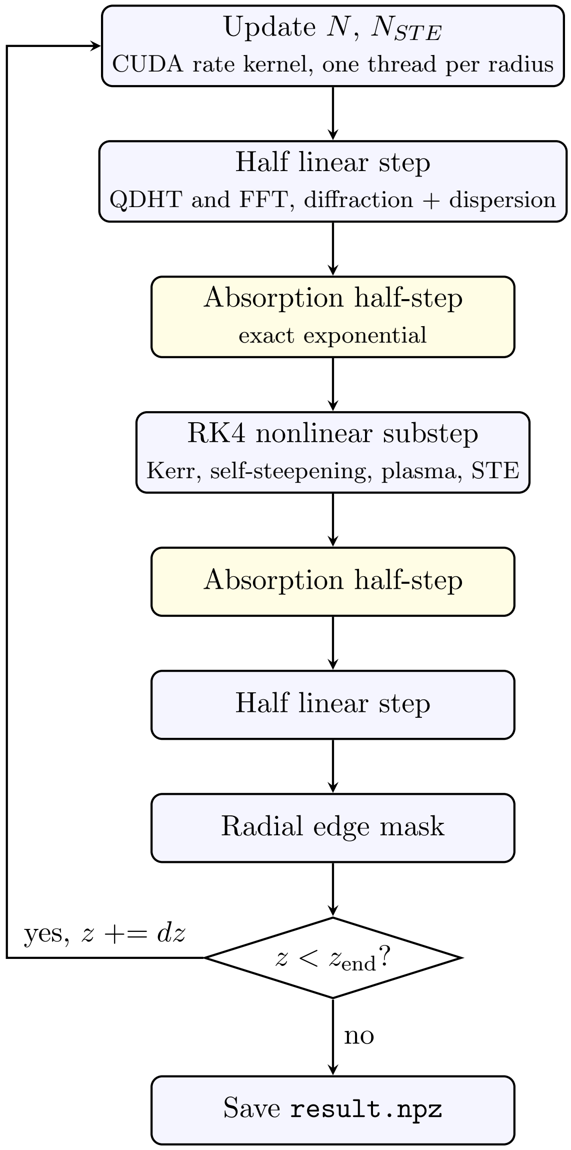


Figure 13: One iteration of the main propagation loop in `Integrator.propagate`. Every z step first advances the carrier populations along the full time axis at fixed r, then applies the symmetric Strang sequence to the field envelope. Diagnostics are recorded every `save_stride` steps, omitted here for clarity, and the loop repeats until z reaches the end of the propagation window.

Note the order of the two blocks at the top. The populations are updated first, using the field from the end of the previous step, and the field is then propagated using those populations held fixed. This is itself a splitting between the field and the carriers, on top of the Strang splitting inside the field update, and it is accurate as long as dz is small enough that neither changes much over one step.

5 GPU Acceleration and CUDA Interpolation

5.1 Why the Grid Needs a GPU

The propagation grid has of order 10^3 radial points, 10^3 to 4×10^3 time points, and 10^4 propagation steps, so the field envelope alone is an array of several million complex numbers that is updated at every step. Every step also needs two matrix multiplications with the QDHT matrix Y, several FFTs along the time axis, and a full evaluation of the nonlinear right hand side at four Runge-Kutta stages. On a CPU this would take hours to days for a single run. The code offloads all of these operations to the GPU through `cupy`, which is a drop in replacement for `numpy` that runs the same array operations on CUDA hardware, so the matrix products, the FFTs and the elementwise nonlinear evaluations all execute in parallel across thousands of GPU cores instead of sequentially. In practice the 4 μJ run used in this report takes about four minutes for 9000 steps.

5.2 The Photoionization Rate Table

The Keldysh photoionization rate W_{PI} derived in Section 3 involves an infinite sum of Dawson functions and complete elliptic integrals. This is far too expensive to evaluate at every grid point and every time step during propagation, since it would dominate the runtime. Instead the code evaluates W_{PI} once, before propagation starts, on a dense logarithmically spaced grid of intensities spanning the full range the simulation can reach, and fits a monotone cubic Hermite interpolant through these points. The result is a lookup table of 4096 points in $\log_{10} I$, stored on the GPU. There are two such tables, one for the valence to conduction band gap U_i and one for the shallower self-trapped exciton gap U_s, which is what makes the re-ionization term of Section 11.2 cheap enough to evaluate at every time sample.

During propagation, whenever the code needs W_{PI} at some intensity, it no longer evaluates the analytical Keldysh formula. It reads the table and interpolates linearly between the two nearest samples in log intensity space. Because the table is sampled logarithmically and the Keldysh rate varies smoothly on a logarithmic intensity scale, this turns a computation involving dozens of transcendental function evaluations into two table reads and one multiplication, at the cost of a small and controlled interpolation error.

5.3 The CUDA Kernel for the Rate Equations

The interpolation above is not only used from the array level. It is also compiled directly into a custom CUDA kernel, written as a raw CUDA C string in `sim/kernels.py` and launched through `cupy.RawKernel`. This is a deliberate departure from the array based style used everywhere else in the code, and it exists because the rate equations of Section 11.2 have to be integrated sequentially in time at every radius. One time sample depends on the previous one, so this step cannot be vectorized as a single array operation the way the FFTs and the QDHT product can.

The kernel assigns one CUDA thread to each radial point. Each thread then walks the entire time axis in a tight loop, from the first sample to the last, evaluating the interpolated rate at every step and advancing N and N_{STE} with the exact exponential update described in Section 12.3. Because thousands of radial points are independent of one another, thousands of these sequential time loops run at the same time on the GPU, one per thread. This is the usual way to make an inherently sequential recurrence fast, which is to parallelize across the dimension that has no sequential dependency, here the radius, instead of trying to parallelize the one that does, here the time.

Two details of the kernel are worth pointing out because they protect the physics rather than the performance. The first sample of every radial line is reset to zero before the loop starts, so each propagation step begins from an unexcited medium at the front of the temporal window instead of inheriting a leftover value. The second is the rescaling applied when $N + N_{STE}$ exceeds N_{at}, which keeps the two populations from together claiming more sites than the material has. Both are cheap guards inside the inner loop, and both matter near the clamping intensity where the depletion factor is doing real work.

5.4 Summary of the GPU Division of Labour

Three different accelerations coexist in the code, each solving a different bottleneck. The QDHT matrix product and the FFTs are handled by the built in `cupy` linear algebra and FFT routines, which are already GPU native and need no custom code. The photoionization rate is precomputed once on the CPU with `scipy`, then transferred to the GPU as a lookup table, which trades an expensive repeated computation for a cheap repeated memory read. The population rate equations, which are sequential in time and cannot be written as a single array operation, are handled by the hand written CUDA kernel that exploits parallelism across the radial coordinate instead. Together these three choices are what make it possible to run a full simulation in two transverse dimensions plus time, over tens of thousands of propagation steps, in minutes rather than days.

6 Interferometric Diagnostics and the Abel Transform

The propagation code produces a three dimensional map of the free electron density, the self-trapped exciton density, and the intensity, but none of these quantities can be measured directly inside a bulk sample. What can be measured is the phase shift accumulated by a probe beam that crosses the excited region, using a Nomarski interferometer. This section derives how that phase shift relates to the refractive index change predicted by the simulation, and describes how `web/abel_phase_explorer.py` implements both directions of this link, from a measured interferogram back to a refractive index profile, and from a simulated refractive index profile forward to a predicted interferogram. The physical setup and the mathematics of the Abel transform below are identical whether the excited medium is air or fused silica, only the propagation equation that produces the refractive index change differs, and that equation was already given in Section 11 for SiO_2.

6.1 Two-Plane-Wave Interferometry

The Nomarski interferometer can be understood as a configuration that allows retrieval of phase information about a medium under investigation. We derive the fundamental equations governing the interference pattern produced by two plane waves with different propagation directions.

6.1.1 Field Configuration and Wave Vectors

Consider two plane waves propagating in the same medium with refractive index η_i:

$$E_1 = \underline{E_0}\, e^{i(\vec{k}_1\cdot\vec{r}-\omega t)} \tag{51}$$

$$E_2 = \underline{E_0}\, e^{i(\vec{k}_2\cdot\vec{r}-\omega t)} \tag{52}$$

where the wave vectors are defined as:

$$\vec{k}_1 = \frac{2\pi}{\lambda}\eta_i \begin{pmatrix}\sin\theta\\ 0\\ \cos\theta\end{pmatrix} = k\begin{pmatrix}\sin\theta\\ 0\\ \cos\theta\end{pmatrix}, \qquad \vec{k}_2 = k\begin{pmatrix}-\sin\theta\\ 0\\ \cos\theta\end{pmatrix} \tag{53}$$

These two plane waves propagate symmetrically with respect to the normal axis, making an angle 2θ with each other. To account for an arbitrary orientation of the fringes around the z axis, we introduce a rotation of angle ϕ, which turns the wave vectors into $k(\cos\phi\sin\theta, \sin\phi\sin\theta, \cos\theta)$ and its mirror image.

6.1.2 Intensity Distribution and Fringe Spacing

Setting the observation screen at $z = 0$, the intensity is:

$$I(x, y) = I_1 + I_2 + \sqrt{I_1 I_2} \cos\left((\vec{k}_1 - \vec{k}_2) \cdot \vec{r}\right) \tag{54}$$

For equal amplitudes and after computing the phase difference $(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} = 2k \sin\theta (x \cos\phi + y \sin\phi)$, this becomes:

$$\boxed{I(x, y) = 2I_0(x, y) \left[1 + \cos\left(2\pi\nu_0(x \cos\phi + y \sin\phi)\right)\right]}, \qquad \nu_0 = \frac{2 \sin\theta}{\lambda} \tag{55}$$

where ν_0 is the spatial-carrier frequency of the fringes, set purely by the crossing angle 2θ of the two interfering beams, and independent of anything happening inside the medium.

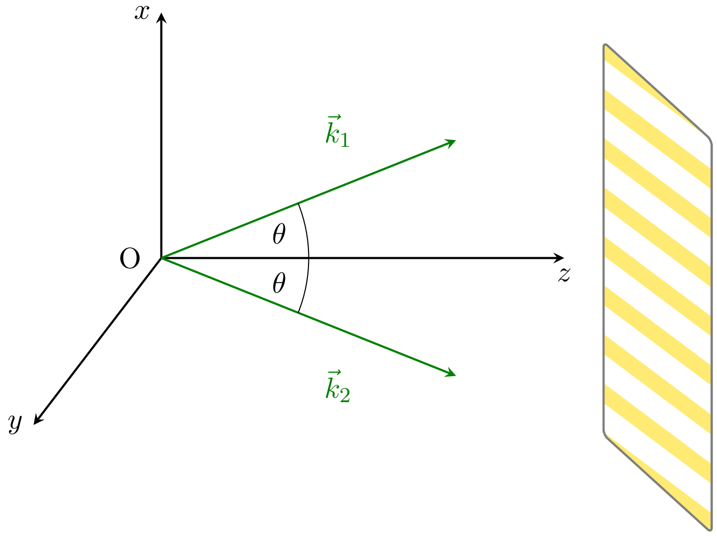


Figure 14: Interference of two plane waves propagating symmetrically with an angle 2θ, creating an interference pattern on the screen (shown for $\phi = 0$).

6.2 Phase and Amplitude Extraction via Fourier Analysis

Nomarski interferograms read $I(x, y) = a(x, y) + b(x, y) \cos\left[2\pi\nu_0 x + \varphi(x, y)\right]$, where $a(x, y)$ is the background term, $b(x, y)$ the amplitude modulation term, and $\varphi(x, y)$ the phase of interest, related to the line integrated electron density by $\varphi(x, y) \propto \int n_e(x, y, z)\, \mathrm{d}z$. In conventional contour-fringe interferometry, obtained when $\nu_0 = 0$, the intensity reads $I = a + b \cos\varphi$, in which a, b, and φ are mixed, which limits accuracy, and the parity of the cosine prevents determining the sign of φ.

The Takeda method [16] overcomes both limitations through spatial homodyne demodulation. As in classical heterodyne detection, the signal of interest is carried by an oscillating reference, here the spatial carrier ν_0 generated by the tilted incidence angle of the interferometer arms. Writing the cosine with Euler's formula gives $I(x, y) = a(x, y) + c(x, y)e^{i2\pi\nu_0 x} + c^*(x, y)e^{-i2\pi\nu_0 x}$ with $c(x, y) = \frac{1}{2} b(x, y)e^{i\varphi(x,y)}$. Bandpass filtering around $+\nu_0$ in the Fourier plane isolates the sideband carrying $c(x, y)$ alone, from which b and φ are recovered respectively as the modulus and the argument.

This linear separation also improves sensitivity to weak phase variations. Contour interferometry responds quadratically to a small phase, $\Delta I_{contour} \approx \frac{b}{2}\varphi^2$, because small variations sit near the extremum of a cosine, whereas the Takeda method responds linearly, $\Delta I_{Takeda} \approx -b \sin(2\pi\nu_0 x)\, \varphi$, which is what gives it sub-wavelength phase sensitivity [16].

Applied to a reference interferogram (no plasma or no excitation) and a measurement interferogram, the ratio of the two filtered and inverse transformed signals cancels the carrier and any residual optical misalignment:

$$z(x,y) \equiv \frac{I_{\text{meas,filt}}(x,y)}{I_{\text{ref,filt}}(x,y)} = \frac{b_2(x,y)}{b_1(x,y)}\, e^{i\varphi(x,y)}, \qquad \boxed{\varphi(x,y) = \text{atan2}\left(\text{Im}\{(z)\}, \text{Re}\{(z)\}\right)} \tag{56}$$

6.3 Resolution Limits

Two independent limits set the finest feature that can be resolved. The Nyquist limit comes from the pixelated sensor. The shortest oscillation observable without aliasing must span at least two pixels, giving $\nu_N = 1/(2\Delta x)$ with Δx the pixel size. The optical limit comes from diffraction at the sample itself. A feature of size d diffracts light at an angle $\sin\theta = \lambda/d$, as illustrated in Figure 16, and this diffracted light is only collected by the imaging lens if $\sin\theta \leq NA$.

6.3.1 Contrast at the Nyquist Limit

The Nyquist frequency $\nu_N = 1/(2\Delta x)$ only says that a fringe period of exactly two pixels can be represented without aliasing. It says nothing about how much contrast survives once the continuous fringe pattern is integrated over the finite area of each pixel, which is what a real sensor does. For fringes tilted at 45°, the intensity recorded by the pixel at (n_x, n_y) is the average of the continuous pattern $\cos^2(2\pi f_x x + 2\pi f_y y + \phi)$ over that pixel:

$$\begin{aligned} I(n_x,n_y) &= \int_{n_x-\frac{1}{2}}^{n_x+\frac{1}{2}} \int_{n_y-\frac{1}{2}}^{n_y+\frac{1}{2}} \cos^2(2\pi f_x x + 2\pi f_y y + \phi)\, dx\, dy \\ &= \frac{1}{2} + \frac{1}{2}\cos(2\pi f_x n_x + 2\pi f_y n_y + \phi) \left[\frac{\sin(\pi f_x)}{\pi f_x}\right] \left[\frac{\sin(\pi f_y)}{\pi f_y}\right] \end{aligned} \tag{57}$$

At the Nyquist limit itself, $f_x = f_y = \frac{1}{2}$, this becomes:

$$I(n_x,n_y) = \frac{1}{2} + \frac{2}{\pi^2}\cos\left(\pi(n_x+n_y) + \phi\right) \tag{58}$$

so successive pixels alternate between $\frac{1}{2} + \frac{2}{\pi^2}$ and $\frac{1}{2} - \frac{2}{\pi^2}$, a modulation of only about ± 0.20 around the mean instead of the full ± 1 of the underlying fringe. Sampling exactly at ν_N is therefore technically alias free but already badly starved of contrast, which is why in practice the sensor should be operated with some margin below ν_N rather than pushed all the way to it.

6.3.2 Separation of Interference Orders in Fourier Space

Filtering a single sideband, as in the Takeda method above, only works if that sideband does not overlap the other two lobes present in the Fourier transform of a recorded interferogram, the zero order at the origin and the conjugate sideband on the other side. The object beam itself has a finite bandwidth set by the optical resolution: for a sensor image magnified by M_{tot}, the smallest resolvable feature is $R_s = M_{tot}\lambda/(2NA)$ by the Abbe criterion, so the object spectrum $\hat{c}(\nu_x, \nu_y)$ is confined to a disk of radius

$$r_f = \frac{1}{2R_s} = \frac{NA}{M_{tot}\lambda}. \tag{59}$$

The zero order term is proportional to $|c|^2$, whose Fourier transform is the autocorrelation $\hat{c} \circledast \hat{c}^*$, so it occupies the larger disk of radius $2r_f$ centered at the origin. The two sidebands sit at $\pm\nu_0$, with ν_0 the carrier frequency of Eq. (55), each carrying a copy of the object spectrum of radius r_f. Center-to-center, non-overlap of the sideband at ν_0 with the central autocorrelation blob requires the separation to exceed the sum of the two radii, giving the same order of magnitude condition as the general one-dimensional argument for off-axis holography, $\nu_0 \gtrsim 3r_f$. Kimura's more detailed two-dimensional treatment of the 45°-tilted configuration used in this setup refines this into two explicit design conditions,

$$\sqrt{3}\,r_f < \nu_0, \qquad \nu_0 + \sqrt{2}\,r_f < \sqrt{2}\,\nu_N, \tag{60}$$

the first ensuring the sideband and the zero order stay separated, the second ensuring the sideband, once separated, still fits entirely inside the region the sensor can sample without aliasing. If the carrier frequency ν_0 is set too low for a given optical resolution r_f, the sideband bleeds into the zero order and no filter can cleanly separate them again, which caps how much optical resolution a given fringe spacing can support, and conversely how tight the fringe spacing must be pushed to preserve a given resolution.

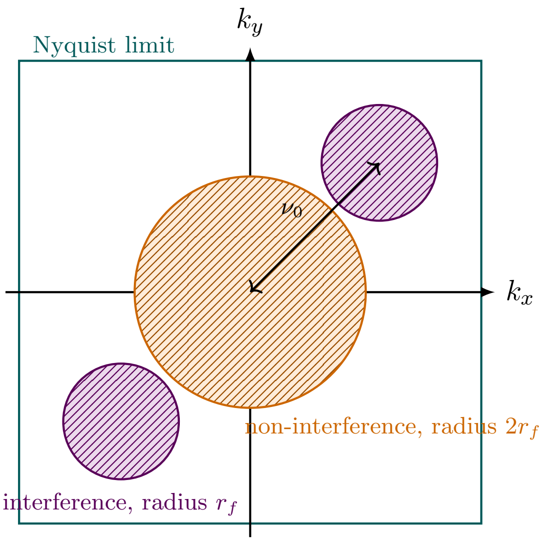


Figure 15: Layout of the interference orders in the Fourier plane of a recorded interferogram. The non-interference (zero order) blob of radius $2r_f$ sits at the origin, while the two interference sidebands of radius r_f sit at $\pm\nu_0$ along the carrier direction. Clean separation for filtering requires both sideband disks to stay outside the zero order disk and inside the square region the sensor can sample without aliasing, set by the Nyquist frequency ν_N.

Inverting the problem gives the smallest resolvable feature for a lens of numerical aperture $NA = \sin\alpha$ in a coherent on axis configuration, $R_0 = \lambda/NA$, shown in Figure 17. After magnification by the imaging system, the highest spatial frequency actually delivered to the sensor is $\nu_{max} = NA/(M_{tot}\lambda)$, and the digital sampling must satisfy $\nu_N \geq \nu_{max}$ for the optical resolution to be preserved rather than lost to undersampling.

6.3.3 Sub-Pixel Correction of the Carrier Frequency

The Fourier transform method above assumes the carrier peak used to recenter the sideband, (u_0, v_0), is known exactly. In practice it is only known to the frequency resolution of the discrete Fourier transform. For a hologram sampled on an $N \times N$ grid, that resolution is $\Delta f_x = \Delta f_y = 1/N$, and nothing guarantees that the true carrier frequency falls exactly on a grid point [19]. If the true peak is offset from the nearest FFT bin by a fractional amount $(\Delta x, \Delta y)$, with $|\Delta x|, |\Delta y| < 0.5$ pixel in frequency space, shifting the sideband to that nearest

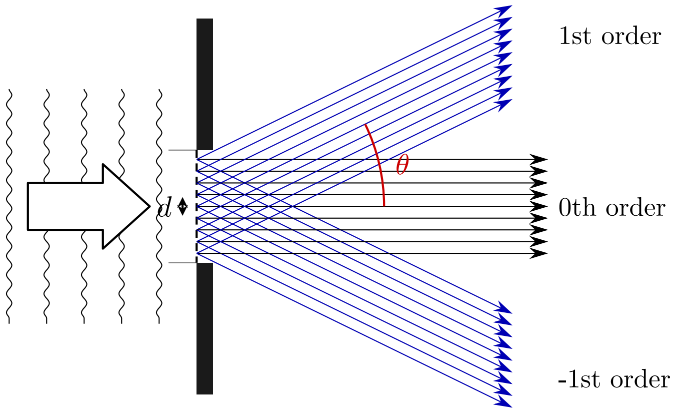


Figure 16: A feature of size d diffracts the probe beam into orders separated by $\sin\theta = \lambda/d$. Only orders within the lens acceptance angle are collected.

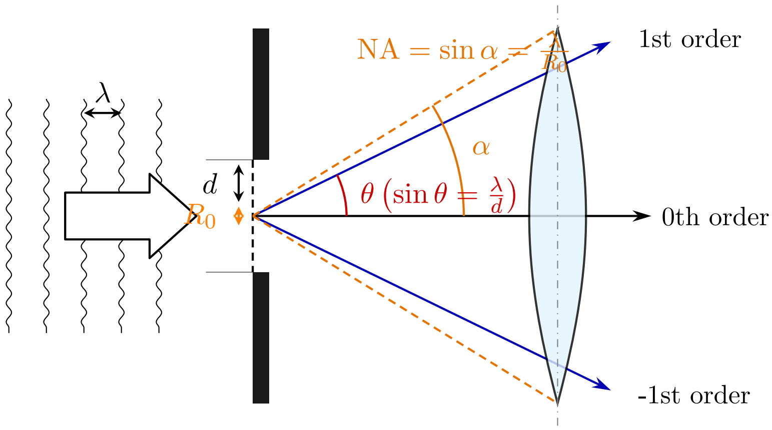


Figure 17: Smallest resolvable feature $R_0 = \lambda/NA$ for a lens of numerical aperture $NA = \sin\alpha$, obtained by requiring the first diffraction order to fall within the acceptance cone.

bin instead of the true peak leaves a residual linear phase ramp across the whole reconstructed phase map:

$$\Delta\varphi(x, y) = \frac{2\pi}{N}\big(\Delta x\, x + \Delta y\, y\big). \tag{61}$$

This shows up as a uniform tilt superimposed on the recovered phase, easy to mistake for a real gradient in the plasma or the sample if left uncorrected.

The fix is to refine the peak location to sub-pixel accuracy before using it, by evaluating a local, up-sampled discrete Fourier transform in a small neighborhood around the coarse FFT maximum rather than trusting the coarse grid point itself. Writing the hologram as $H(X, Y)$ on the spatial grid $X = Y = [-N/2, \dots, N/2-1]^T$, this local transform is the matrix product

$$F(U, V) = \exp\Big(-i\tfrac{2\pi}{N}UX^T\Big)\, H(X, Y)\, \exp\Big(-i\tfrac{2\pi}{N}YV^T\Big), \tag{62}$$

evaluated only on a small set of frequencies U, V centered on the coarse peak (u_0, v_0) and spaced by $1/\alpha$ instead of 1, where $\alpha \sim 10$ is an up-sampling factor [19]. Because U and V only span a window of a couple of pixels around (u_0, v_0), this costs little compared with the full $N \times N$ transform even though it resolves the peak far more finely. The maximum of $|F(U, V)|$ in this window gives the refined peak $(u_0 + \Delta x, v_0 + \Delta y)$, from which $\Delta\varphi(x, y)$ above is computed and subtracted from the phase map obtained by the ordinary four step Fourier transform method, removing the spurious tilt without needing a higher resolution FFT of the entire hologram.

This correction is not currently applied in the analysis code. `abel_phase_explorer.py` locates the carrier peak with a single coarse FFT maximum in `find_1storder_peak`, and relies on `fit_plane_from_mask` to remove any resulting tilt from a background region afterwards. That corrects the same symptom, but only where a flat background region is actually available in the image, so refining the peak at the peak finding step would be the more robust fix.

6.4 The Abel Transform

Consider a probe beam crossing the excited cylindrically symmetric region along y, at fixed x and z. The accumulated phase relative to a beam that has not crossed the excited region is:

$$\phi(x,z) = 2\left(\frac{2\pi}{\lambda_L}\right)\int_0^\infty [\eta(r,z)-1]\,dy, \qquad r=\sqrt{x^2+y^2} \tag{63}$$

Using the substitution $y=\sqrt{r^2-x^2}$, this line integral becomes the forward Abel transform of the refractive index deviation $f(r)\equiv\eta(r,z)-1$:

$$\boxed{F(x) = 2\int_x^\infty \frac{f(r)\,r}{\sqrt{r^2-x^2}}\,dr}, \qquad F(x)\equiv\frac{\lambda_L}{4\pi}\phi(x,z) \tag{64}$$

This is consistent with equation (4) of reference [18]. The Abel transform can be inverted analytically. Applying the operator $\int_\rho^\infty x\,dx/\sqrt{x^2-\rho^2}$ to both sides and using the identity $\int_\rho^r x\,dx/\sqrt{(r^2-x^2)(x^2-\rho^2)}=\pi/2$ gives, after differentiating with respect to ρ:

$$\boxed{f(\rho) = -\frac{1}{\pi}\int_\rho^\infty \frac{dF(x)/dx}{\sqrt{x^2-\rho^2}}\,dx} \qquad \Longrightarrow \qquad \boxed{\eta(r,z) = 1-\frac{1}{\pi}\left(\frac{\lambda_L}{2\pi}\right)\int_r^\infty \frac{d\phi(x,z)/dx}{\sqrt{x^2-r^2}}\,dx} \tag{65}$$

The Abel transform is also a special case of the more general Radon transform, which describes the line integral of a function along an arbitrary ray at angle θ and impact parameter ρ. For a rotationally symmetric function $f(r)$, the projection is the same at every angle, and choosing $\theta=0$ turns the Radon integral into $\mathcal{R}[f](\rho)=2\int_\rho^\infty f(r)\,r\,dr/\sqrt{r^2-\rho^2}$, exactly the forward Abel transform above. This equivalence is what justifies treating the interferometric phase, a line integrated quantity, as a projection that can be inverted whenever cylindrical symmetry holds.

6.5 Numerical Inversion by the Trapezoidal Rule

To invert the Abel transform numerically, the integral is approximated by summing trapezoidal areas over the integrand g,

$$A_k \approx \tfrac{1}{2}(x_{k+1}-x_k)\big[g(x_{k+1})+g(x_k)\big], \qquad g(x_k,r_i) = \frac{\big[dF(x)/dx\big]_{x=x_k}}{\sqrt{x_k^2-r_i^2}}. \tag{66}$$

The derivative is itself approximated by a forward difference, $dF/dx|_{x_k}\approx[F(x_{k+1})-F(x_k)]/\Delta x$, and the integration starts one pixel away from $x=r_i$ to avoid the mathematical singularity there. Substituting the forward difference into the trapezoidal sum, the step size Δx cancels exactly, leaving:

$$\boxed{f(r_i) = -\frac{1}{2\pi}\sum_{k=i+1}^{N-2}\left[\frac{F(x_{k+2})-F(x_{k+1})}{\sqrt{x_{k+1}^2-r_i^2}}+\frac{F(x_{k+1})-F(x_k)}{\sqrt{x_k^2-r_i^2}}\right]} \tag{67}$$

directly implementable without ever storing Δx as a separate factor.

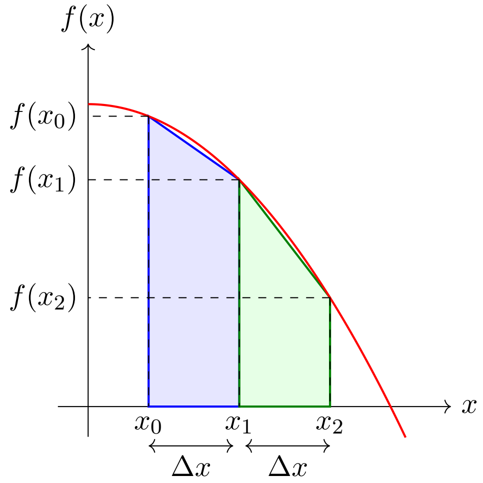


Figure 18: Illustration of the trapezoidal rule for numerical integration. The integral under the curve is approximated as a sum of trapezoidal areas.

6.6 Connecting the Simulation to the Diagnostic

The script `web/abel_phase_explorer.py` implements both directions of the link between the simulation and a real interferometric measurement, so that the two can be compared directly.

On the experimental side, `find_1storder_peak` locates the sideband of a raw interferogram in Fourier space by masking out the central DC region and searching for the strongest remaining peak, which is the Takeda demodulation step described above. `reconstruct` then crops a disk of radius set by the numerical aperture around that peak and inverse transforms it back to real space, giving the complex field $c(x, y)$ from which the phase is extracted. `fit_plane_from_mask`, applied over background regions selected through `baseline_rects_to_mask`, removes any residual linear phase tilt left by small optical misalignments that dividing by the reference does not fully cancel. `preprocess_side_full` chains these steps into the full pipeline, from a raw pair of interferograms, signal and reference, to a clean two dimensional phase map.

On the simulation side, `build_abel_matrix` discretizes the forward Abel transform directly as a matrix acting on the non uniform radial grid produced by the QDHT. It uses the exact shell integral of $r/\sqrt{r^2 - x^2}$ over each radial bin rather than a trapezoidal sum on that grid, so that the same relation $F(x) = 2\int_x^{\infty} f(r)r\,dr/\sqrt{r^2 - x^2}$ used above for the inversion is applied here in the forward direction. `channel_phases_2d` then assembles the refractive index change and turns it into a phase map.

6.7 The Dielectric Function Seen by the Probe

Everything above is about extracting a phase from fringes. What that phase means still has to be decided, and this is where the probe analysis departs from the pump side model of Section 11. The pump only ever needs one number, the index at its own wavelength, and Section 11.4 gave it as a single bound oscillator. The probe is a different beam at a different wavelength, and it measures two things at once, a phase shift and a change in transmission. Both have to come from the same physics or the two panels of the diagnostic will disagree with each other.

The model used here is the dielectric function of Martin et al. [6], who measured exactly this kind of pump probe phase shift in wide gap crystals and had to build an index model to interpret it. Their Eq. (2) writes the permittivity of the excited medium as the unexcited permittivity plus one term per population,

$$\varepsilon_2(\omega) = \varepsilon_1(\omega) \underbrace{-\frac{f_{CB}\,\omega_{p,CB}^2}{\omega^2 + i\omega/\tau_{ep}}}_{\text{free electrons, Drude}} + \underbrace{\sum_j \frac{f_j\,\omega_{p,tr}^2}{\omega_j^2 - \omega^2 - i\omega\gamma_j}}_{\text{trapped electrons, Lorentz}} \underbrace{-\,(n_0^2 - 1)\,\frac{N + N_{STE}}{N_{val}}}_{\text{valence depletion}} + \underbrace{2n_0\,X\,n_2 I}_{\text{cross Kerr}} \tag{68}$$

where $\omega_p^2 = Ne^2/(\varepsilon_0 m)$ is the plasma frequency of the population concerned. Each term deserves a word, because three of them are absent from the simpler treatment of Section 11.4.

The first term is the free electron response, and it is the one that makes the phase go negative. It is written as a Drude term with a collision time τ_{ep} rather than as the collisionless $-N/2\rho_c$ used earlier in this report. Keeping the collision term matters because it moves weight from the real part into the imaginary part, which is precisely the absorption the transmittance panel shows. The effective mass is $0.5\,m_e$ rather than the bare mass, following Table II of the same paper.

The second term is the trapped population, and it is the one that brings the phase back up. It is a sum of Lorentz oscillators rather than the single one of Section 11.4, because the STE absorption in fused silica is not one clean line. Martin et al. fit two bands, one at 5.2 eV with oscillator strength 0.40 and width 1.5 eV, and one at 4.2 eV with strength 0.15 and width 1.0 eV. The mass here is the bare electron mass, not the effective mass, because a trapped electron is bound and no longer moves through the band. Using one band with unit oscillator strength instead of these two overestimates the STE contribution at 515 nm by a factor of about 2.8, which is large enough to change the sign of the late time plateau.

The third term is the one most easily forgotten. Every electron promoted to the conduction band, or subsequently trapped, is an electron that has left the valence band, where it was contributing to the ordinary refractive index of the unexcited glass. Removing it removes that contribution. The term is negative, proportional to the total number of electrons removed, and normalised by N_{val}, the density of valence electrons. That normalisation is worth stating carefully, since the same symbol N_0 in Table II of the paper is used for the density of ionisable units, about 2.2×10^{22} cm^{-3} for SiO_2, whereas what the depletion term needs is the number of electrons carrying the polarisability, which is eight per SiO_2 unit for the four Si-O bonds. Using the unit density instead of the electron density makes this term eight times too strong, which is enough to flip the long delay plateau from positive to negative.

The fourth term is the Kerr response of the glass to the pump intensity, felt by the probe. It carries a factor X that is not present when a beam acts on itself. A weak probe at a different frequency sees twice the nonlinear index that the pump sees from its own intensity, because the cross term in the third order susceptibility is counted twice. Whether X is 1 or 2 therefore depends on how n_2 was defined, and the code exposes it as `xpm_factor` rather than hiding the choice.

From this single complex permittivity the two observables follow directly. Writing ε_1 for the unexcited medium and ε_2 for the excited one,

$$\Delta n = \mathrm{Re}\,\sqrt{\varepsilon_2} - \mathrm{Re}\,\sqrt{\varepsilon_1}, \qquad \alpha = \frac{2\omega}{c}\,\mathrm{Im}\,\sqrt{\varepsilon_2}, \tag{69}$$

which are Eqs. (3) and (4) of the same paper. Taking the square root rather than the usual first order expansion $\Delta n \simeq \Delta\varepsilon/2n_0$ is not a detail at these densities. The expansion assumes the excited medium is a small perturbation of the unexcited one, and near the clamping density that assumption fails. In the extreme case, where Re ε_2 turns negative, the expansion still returns a large negative index while the exact expression correctly says the medium has stopped transmitting and started reflecting. The code flags that condition rather than reporting a phase for it.

The practical gain is that the phase panel and the transmittance panel of the diagnostic are now two views of one calculation. The phase map is the Abel transform of Δn, and the transmittance map is $\exp\left(-\int \alpha\, \mathrm{d}x\right)$ along the same line of sight, both taken from the same ε_2. Under the older treatment, where the phase came from a hand written index formula and the absorption from a separately chosen cross section, nothing forced the two to be consistent, and they were not.

Each contribution is Abel transformed and low pass filtered separately, one channel at a time. This works because both operations are linear, so summing the channels commutes with both the line of sight integral and the filter. The reason for doing it this way is practical, since the browser can then switch individual channels on and off by summing whichever ones are ticked, with no transform to redo. Three channels are exposed, namely the free electron response which is negative, the trapped exciton response which is positive, and the cross Kerr response which is also positive. The valence depletion term is folded into the free electron channel, because both scale with the number of electrons that ionization has removed from the valence band.

6.7.1 Filtering by the Numerical Aperture

One step separates the predicted phase map from a raw line integral. The simulated Δn contains spatial structure finer than the collection optics can transmit, so comparing it directly to a measured map would compare things that were never measured the same way. `lowpass_NA_2d` therefore applies the resolution limit derived earlier in this section, by Fourier transforming the map in the (z, x) plane, zeroing everything outside the disk of radius $2\pi NA/\lambda$, and transforming back. This is the same r_f cutoff that set the order separation conditions above, applied here as a forward model of what the imaging system actually delivers to the sensor.

6.7.2 Choosing the Delay and the Time Origin

The probe crosses the sample at one instant, but the pump takes time to travel through it, so the pump slice a probe ray meets depends on where along z that ray crosses. For a probe delay τ, the code evaluates the local time

$$t_{local}(z) = \tau - \frac{z - z_{ref}}{v_g}, \tag{70}$$

with v_g the pump group velocity from the Sellmeier group index, and picks the nearest sample of the stored time axis. The reference plane z_{ref} is what defines $\tau = 0$. Setting it at the entrance face means $\tau = 0$ is the moment the pump enters the medium. Setting it at the collapse point, which is what `auto_t0_ref_um` finds by locating the maximum on axis intensity, means $\tau = 0$ is the coincidence of the pump and probe maxima, which is the convention used in pump probe measurements. The distinction is not cosmetic. In a 350 μm box the pump needs a little over a picosecond to cross the sample, which is longer than the whole delay range of interest, so the two conventions place the Kerr peak at completely different delays.

6.7.3 The Interactive Output

The result of all of this is a self contained HTML page, one per probe wavelength, generated by `build_explorer_html`. It contains the phase maps for every delay in the scan, already Abel transformed and NA filtered, so no computation happens in the browser beyond summing channels. A slider steps through the delays. Tick boxes switch the three Δn channels on and off, which is how the contribution of each mechanism to the total phase is read off directly rather than inferred. The page shows the phase map in the (z, x) plane, the on axis phase profile along z, the transmittance map from the imaginary part of the same permittivity, and a separate panel with the electron and STE densities in the (z, r) plane at the same delay.

That last panel needs one word of caution. The density maps are the real densities in the meridian plane as the solver computed them, not line of sight integrals, so they are not what a camera would record. They are there to explain the phase maps above them, since the whole point of the exercise is that the measured phase is an integral through a structure the measurement cannot resolve directly. The phase panels are the ones that can be compared with an experiment.

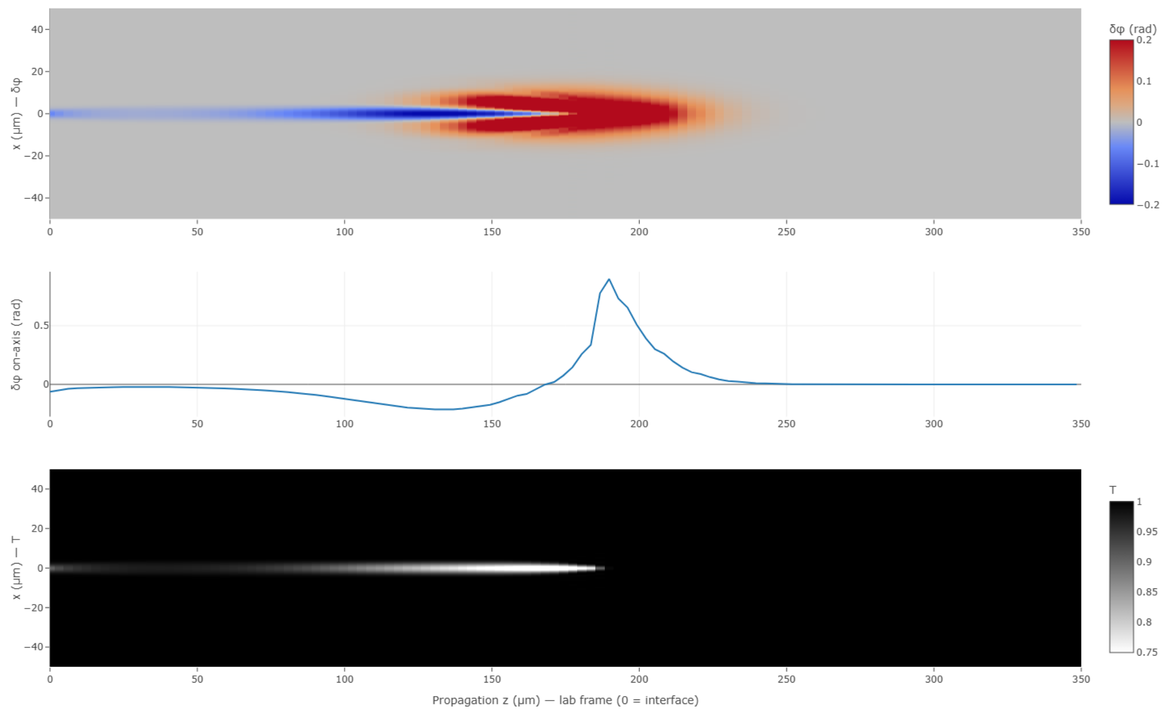


Figure 19: What the diagnostic page shows for the 4 μJ run at one probe delay. Top, the phase map $\delta\varphi(z, x)$ after the forward Abel transform and the numerical aperture filter, with the three Δn channels summed. Middle, the same phase read along the axis. Bottom, the transmittance over the same plane, obtained from the imaginary part of the same permittivity. The phase is negative before the nonlinear focus, where free electrons dominate, and turns strongly positive around $z \simeq 190$ μm where the Kerr and trapped exciton contributions take over. The transmittance dips only along the thin channel where the electron density is high, which is the absorption counterpart of the negative phase region above it.

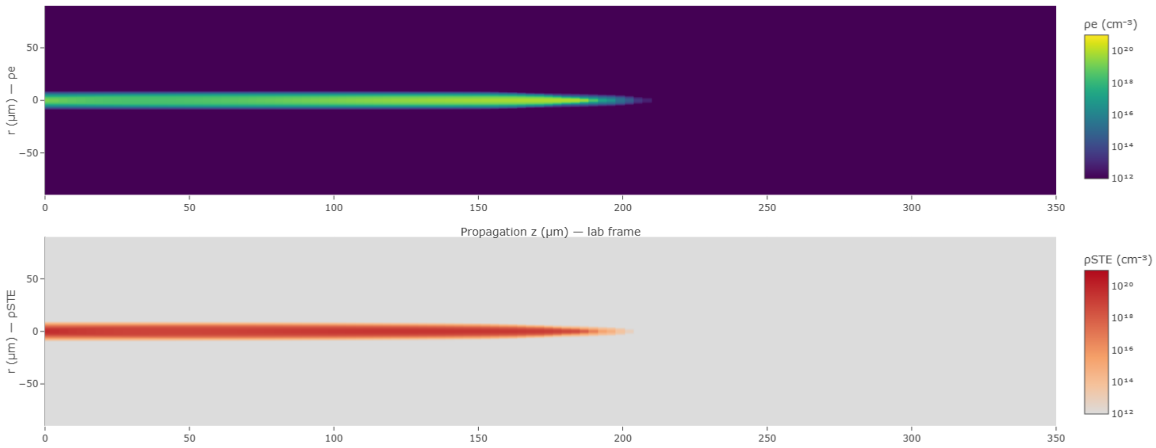


Figure 20: Free electron density (top) and self-trapped exciton density (bottom) in the meridian plane at the same delay as Figure 19, on a logarithmic scale. These are the densities the solver computed, not line of sight integrals, so they are not directly measurable. They are shown because they explain the phase maps above. The electron channel is narrow and stops around $z \simeq 210$ μm where the pulse has been depleted, while the trapped population extends over the same region and survives after the electrons have gone, which is what produces the positive phase at late delays.

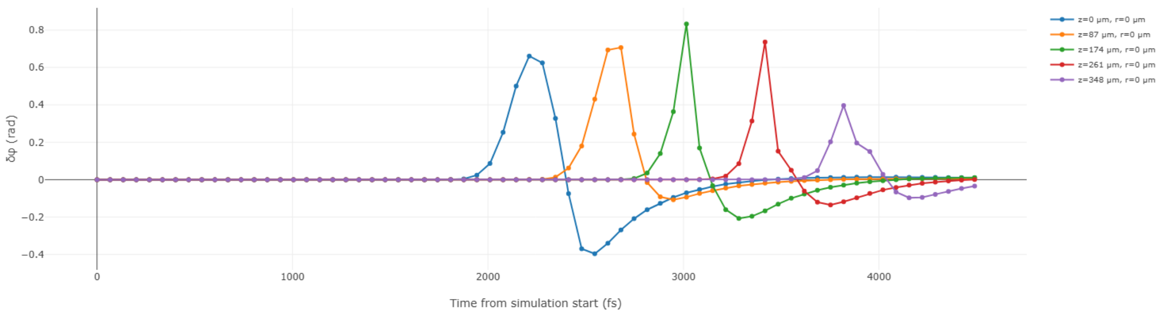


Figure 21: On-axis probe phase shift as a function of time, at five planes along the propagation. Each curve shows the same three stage sequence, a positive peak while the pump is present, a swing to negative once free electrons appear, then a slow recovery toward zero as trapping and decay drain the population. The five peaks are staggered in time because the pump needs about 1.7 ps to cross the 350 μm box, so a plane further along is reached later. This staggering is the reason the time origin has to be stated explicitly, as discussed above. The peak is largest near $z = 174$ μm, close to the nonlinear focus, and the negative excursion is deepest at the entrance plane where the electron density builds up without the compensating Kerr contribution of the focus.

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A Algorithm Listings

The pseudocode below gives the exact, line by line correspondence with `NewSim3juillet.py` referenced throughout Sections 11 to 13.

Subroutine SELLMEIER *(Fused-Silica Linear Refractive Index)***:**

Input: Vacuum wavelength λ [m]

Output: Linear refractive index $n(\lambda)$

$\lambda_{\mu m} \leftarrow \lambda \times 10^6$ `// Convert to micrometres for tabulated coefficients`

$n^2 - 1 \leftarrow \sum_{i=1}^{3} \dfrac{B_i\,\lambda_{\mu m}^2}{\lambda_{\mu m}^2 - L_i^2}$ `// Three-term fused-silica Sellmeier sum`

return $n \leftarrow \sqrt{1 + (n^2 - 1)}$ `//` $\{B_i\}, \{L_i\}$`: fused-silica constants`

Algorithm 1: Linear index of fused silica (`n_sellmeier`).

Class CONFIG *(Simulation Parameters & Derived Physical Constants)***:**

Data: Default attributes grouped by role: numerical grid (`nz`, `Nt`, `N`, ...); laser (λ, `energy_uJ`, `w0`, `delta_t`); material SiO_2 (`n2`, `Ui_eV`, `tau_c`, ...); STE channel (`Us_eV`); physics switches; probe (`lambda_probe`); adaptive-step controls

Method POSTINIT *(`__post_init__`): derive constants & grid quantities***:**

Output: Instance augmented with derived physical constants and grid scalars

1. Angular Frequencies & Probe Optics:

$\omega_0 \leftarrow 2\pi c/\texttt{wavelength}$ // Pump carrier angular frequency

$\omega_{pr} \leftarrow 2\pi c/\texttt{lambda\_probe}$ // Probe angular frequency

$n_{0,pr} \leftarrow \textsc{Sellmeier}(\texttt{lambda\_probe})$ // Probe linear index

$\rho_{c,pr} \leftarrow (\varepsilon_0 m_e \omega_{pr}^2/e^2) \times 10^{-6}$ // Probe critical plasma density [cm^{-3}]

2. Pump Linear Optics (Sellmeier):

$n_0 \leftarrow \textsc{Sellmeier}(\texttt{wavelength})$ // Pump linear index

$k_0 \leftarrow (2\pi/\texttt{wavelength})\, n_0$ // Pump wavenumber

3. Energy Gaps & Effective Masses:

$U_i \leftarrow \texttt{Ui\_eV} \cdot e$ // Valence bandgap [J]

$U_s \leftarrow \texttt{Us\_eV} \cdot e$ // STE gap [J]

$m_{eff} \leftarrow \texttt{meff\_rel} \cdot m_e$ // Reduced effective mass (Keldysh)

$m_D \leftarrow \texttt{meff\_drude\_rel} \cdot m_e$ // Drude effective mass

4. Kerr & Raman Constants:

$\chi^{(3)} \leftarrow \frac{4}{3}\varepsilon_0 n_0^2 c\, \texttt{n2}$ // Third-order susceptibility from n_2

$\Omega_R^2 \leftarrow 1/\texttt{tau\_s}^2 + 1/\texttt{tau\_d}^2$ // Raman oscillator resonance frequency^2

5. Longitudinal Grid:

$\Delta z \leftarrow (\texttt{end} - \texttt{begin})/\texttt{nz}$ // Uniform propagation step

$b \leftarrow k_0\, \texttt{w0}^2$ // Confocal parameter (twice the Rayleigh range)

if *`dz_init` is unset* **then**

  $\texttt{dz\_init} \leftarrow \min(\Delta z, \texttt{dz\_max})$ // Initial adaptive-step ceiling

end

6. Peak Intensity & Field Amplitude:

if *`peak_power_W` is provided* **then**

  $I_0 \leftarrow 2\,\texttt{peak\_power\_W}/(\pi\, \texttt{w0}^2)$ // Gaussian peak intensity from peak power

else

  $I_0 \leftarrow 2\,\mathcal{E}_p/(\pi\, \texttt{w0}^2\, \texttt{delta\_t})$ // Pulse energy $\mathcal{E}_p = \texttt{energy\_uJ} \cdot 10^{-6}$

end

$I_0^{\mathrm{cm}} \leftarrow I_0 \times 10^{-4}$ // W/m^2 $\rightarrow$ W/cm^2 (Keldysh LUT input)

$E_0 \leftarrow \sqrt{2I_0/(n_0 c\, \varepsilon_0)}$ // Peak electric-field amplitude

7. Radial Sampling Diagnostic:

$\Delta r \leftarrow \texttt{R\_factor}\, \texttt{w0}/(\texttt{N} - 1)$ // Radial grid spacing

$p_{w_0} \leftarrow \texttt{w0}/\Delta r$ // Radial points per beam waist w_0

if $p_{w_0} < 3$ **then**

  warn "radial under-sampling" // QDHT accuracy compromised; raise N

end

else if $p_{w_0} < 6$ **then**

  warn "marginal radial sampling" // Borderline transverse resolution

end

Algorithm 2: Simulation configuration and derived constants (`Config`).

Class KeldyshSiO2 *(Solid-State Keldysh Photoionization Model)*:

Method Constructor *(`__init__`): store parameters & derive constants*:

Input: Wavelength λ; gap `Ui_eV`; effective mass m_{eff}; linear index n_0; series length `N_sum`

Output: Instance holding $\{U, m_{eff}, n_0, \omega, \hbar, \texttt{N\_sum}\}$

$U \leftarrow \texttt{Ui\_eV} \cdot e$ // Bandgap energy [J]

$\omega \leftarrow 2\pi c/\lambda$ // Carrier angular frequency

Store $m_{eff}, n_0, \hbar, \texttt{N\_sum}$ as instance attributes // Direct parameter retention ($\hbar$: SI constant)

Method Rate *(`rate`): analytical photoionization rate*:

Input: Intensity I [W/cm^2]

Output: Photoionization rate $W_{PI}(I)$ [cm^{-3} s^{-1}]

1. Field & Keldysh Parameter:

$I_{SI} \leftarrow I \times 10^4$ // W/cm^2 $\rightarrow$ W/m^2

$E \leftarrow \sqrt{2I_{SI}/(n_0 c\,\varepsilon_0)}$ // Peak field driving the ionization

$\gamma \leftarrow \max\big(\omega\sqrt{m_{eff}U}/(eE),\ 10^{-12}\big)$ // Keldysh parameter; floored to avoid $E \rightarrow 0$ singularity

$\gamma_1 \leftarrow \gamma/\sqrt{1+\gamma^2}$ // Tunneling-side modulus

$\gamma_2 \leftarrow 1/\sqrt{1+\gamma^2}$ // Multiphoton-side modulus ($\gamma_1^2 + \gamma_2^2 = 1$)

2. Complete Elliptic Integrals (Dual Modulus):

$K_1, E_1 \leftarrow \text{EllipticK}(\gamma_1^2),\ \text{EllipticE}(\gamma_1^2)$ // Parameter $m = \gamma_1^2$

$K_2, E_2 \leftarrow \text{EllipticK}(\gamma_2^2),\ \text{EllipticE}(\gamma_2^2)$ // Parameter $m = \gamma_2^2$

3. Effective Ionization Order:

$x \leftarrow 2UE_2/(\pi\gamma_1\hbar\omega)$ // Effective bandgap in dressed-photon units (MPI order)

4. Dawson Series (Keldysh Q-function):

for $n \leftarrow 0$ **to** *N_sum* $- 1$ **do**

$a_n \leftarrow \pi\sqrt{\big(2\lfloor x+1\rfloor - 2x + n\big)/(2K_2E_2)}$ // Argument of the n-th quasi-energy channel

$s_n \leftarrow \exp\big(-\pi n(K_1 - E_1)/E_2\big) \cdot \text{Dawson}(a_n)$ // Channel weight $\times$ Dawson integral

end

$\Sigma \leftarrow \sum_{n=0}^{\texttt{N\_sum}-1} s_n$ // Sum over ionization channels

$Q \leftarrow \sqrt{\pi/(2K_2)}\,\Sigma$ // Keldysh Q-function

5. Rate Assembly:

$\text{pref} \leftarrow \dfrac{2\omega}{9\pi}\Big(\dfrac{\omega m_{eff}}{\hbar\gamma_1}\Big)^{3/2}$ // Keldysh prefactor [m^{-3} s^{-1}]

$W_{PI} \leftarrow \text{pref} \cdot Q \cdot \exp\big(-\pi\lfloor x+1\rfloor(K_1 - E_1)/E_2\big) \times 10^{-6}$ // Exponential MPI/tunnel factor; $\times 10^{-6}$: m^{-3} $\rightarrow$ cm^{-3}

return W_{PI}

Algorithm 3: Solid-state Keldysh photoionization model (`KeldyshSiO2`).

Function InterpolatePIRate *(Log-Linear LUT Lookup of W_{PI})*:

Input: Intensity I; rate table T of size N; log-grid origin $\log I_{min}$; inverse log-step $\Delta_{\log}^{-1}$; bounds I_{min}, I_{max}

Output: Interpolated photoionization rate $W_{PI}(I)$

1. Sanitization & Clamping:

if $\neg\, \mathit{isfinite}(I) \vee I < 0$ **then**

  $I \leftarrow 0$ `// Reject NaN/Inf/negative intensity`

end

$I \leftarrow \mathrm{Clip}(I, I_{min}, I_{max})$ `// Restrict to tabulated range`

2. Fractional Table Index:

$f_{idx} \leftarrow (\log_{10} I - \log I_{min})\, \Delta_{\log}^{-1}$ `// Map onto uniform log-intensity grid`

$idx \leftarrow \mathrm{Clip}(\lfloor f_{idx} \rfloor,\ 0,\ N-2)$ // Keep $idx{+}1$ valid for interpolation

3. Log-Linear Interpolation:

$W_{PI} \leftarrow T[idx] + (f_{idx} - idx)\,(T[idx{+}1] - T[idx])$ `// Linear blend between adjacent samples`

4. Output Sanitization:

if $\neg\, \mathit{isfinite}(W_{PI}) \vee W_{PI} < 0$ **then**

  return 0 `// Guard against invalid rate`

end

return W_{PI}

3. Output Sanitization:

if $\neg\, \mathit{isfinite}(x_{new}) \vee x_{new} < 0$ **then**

  return 0 `// Enforce physical non-negativity`

end

return x_{new}

Algorithm 4: Exact exponential integrator for $dx/dt = S + Lx$, i.e. $x(\Delta t) = x_0 e^{L\Delta t} + S\Delta t\, \varphi_1(L\Delta t)$ with $\varphi_1(z) = (e^z - 1)/z$, Taylor fallback for small $|z|$ (`exact_exp_step`).

Function ExactExpStep *(Exact Exponential Integrator for $\dot{x} = S + Lx$)*:

Input: Population x; source S; linear coefficient L; time step Δt

Output: Updated population x_{new}

1. Exponential φ_1 Function:

```
L_Δ ← L Δt                                      // Dimensionless linear exponent
if |L_Δ| > 10^{-6} then
    φ_1 ← (e^{L_Δ} − 1)/L_Δ                     // Exact φ_1(z) = (e^z − 1)/z
else
    φ_1 ← 1 + L_Δ(1/2 + L_Δ(1/6 + L_Δ/24))      // Taylor fallback, avoids 0/0 at
                                                   small |L_Δ|
end
```

2. Exact Update:

```
x_new ← e^{L_Δ} x + S Δt φ_1                    // Closed-form solution over Δt
if ¬ isfinite(x_new) ∨ x_new < 0 then
    return 0                                    // Enforce physical non-negativity
end
return x_new
```

Algorithm 5: Exact exponential integrator for $dx/dt = S + Lx$, i.e. $x(\Delta t) = x_0 e^{L\Delta t} + S\Delta t\, \varphi_1(L\Delta t)$ with $\varphi_1(z) = (e^z - 1)/z$ (Taylor for small $|z|$).

Function SolveRateEquationKernel *(Part 1/2: CUDA Setup & $t = 0$ Initialization)*:

Input: Field envelope $\mathcal{E}$; density arrays n_e, n_s; grid sizes N_r, N_t; step Δt; field-to-intensity factor κ; avalanche coefficients β_g, β_s; neutral density n_a; recombination rate τ_r^{-1}; STE flag E_{ste}; PI tables T_g (CB), T_s (STE) with shared log-grid parameters

Output: Per-thread registers initialized at $t = 0$; time-stepping loop launched

1. Thread Mapping (one thread → one radial point):

$r_i \leftarrow \text{blockDim.x} \cdot \text{blockIdx.x} + \text{threadIdx.x}$ // Global radial index
if $r_i \geq N_r$ **then**
  return // Early exit for out-of-bound threads
end
$base \leftarrow r_i \cdot N_t$ // Row offset into flattened (r, t) arrays

2. Initial Densities at $t = 0$:

$n_e \leftarrow n_e[base], \quad n_s \leftarrow n_s[base]$ // Load into local registers
if $\neg\, \mathit{isfinite}(n_e) \vee n_e < 0$ **then**
  $n_e \leftarrow 0$
end
if $\neg\, \mathit{isfinite}(n_s) \vee n_s < 0$ **then**
  $n_s \leftarrow 0$ // Sanitize seed densities
end

3. Initial Intensity & PI Rates:

$\mathrm{f}_0 \leftarrow \mathcal{E}[base]$ // Complex field envelope at $t = 0$
$I_c \leftarrow (f_{0,x}^2 + f_{0,y}^2)\,\kappa$ // Intensity $= |\mathcal{E}|^2\kappa$
$W_c \leftarrow \text{InterpolatePIRate}(I_c, T_g, \dots)$ // CB photoionization rate
if $E_{ste} \neq 0$ **then**
  $W_c^s \leftarrow \text{InterpolatePIRate}(I_c, T_s, \dots)$ // STE photoionization rate
else
  $W_c^s \leftarrow 0$ // STE channel disabled
end

4. Time-Stepping Loop:

for $j \leftarrow 0$ **to** $N_t - 2$ **do**
  Execute TimeStepCore (Algorithm 7) // Sequential march over temporal grid
end

5. Next-Step Field, Intensity & PI Rates:

$\mathrm{f}_1 \leftarrow \mathcal{E}[base + j + 1]$ // Envelope at step $j{+}1$
$I_n \leftarrow (f_{1,x}^2 + f_{1,y}^2)\,\kappa$ // Next-step intensity
$W_n \leftarrow \text{InterpolatePIRate}(I_n, T_g, \dots)$ // CB photoionization rate
if $E_{ste} \neq 0$ **then**
  $W_n^s \leftarrow \text{InterpolatePIRate}(I_n, T_s, \dots)$ // STE photoionization rate
else
  $W_n^s \leftarrow 0$ // STE channel disabled
end

Algorithm 6: Two-population plasma rate-equation solver (Part 1/2): thread mapping, validation, $t = 0$ initialization (`solve_rate_equation_kernel`).

Function TimeStepCore *(Part 2/2: Trapezoidal Step $j \to j{+}1$)*:

Input: Thread registers $base, n_e, n_s, I_c, W_c, W_c^s$; step index j

Output: Global memory at $base + j + 1$ written; registers advanced

1. Trapezoidal Averaging over $[j, j{+}1]$:

```
I_avg ← ½(I_c + I_n)                                  // Mean intensity
W_avg ← ½(W_c + W_n)                                  // Mean CB PI rate
W^s_avg ← ½(W^s_c + W^s_n)                            // Mean STE PI rate
```

$I_{avg} \leftarrow \frac{1}{2}(I_c + I_n)$ // Mean intensity

$W_{avg} \leftarrow \frac{1}{2}(W_c + W_n)$ // Mean CB PI rate

$W_{avg}^s \leftarrow \frac{1}{2}(W_c^s + W_n^s)$ // Mean STE PI rate

2. Valence-Band Depletion Factor:

$D \leftarrow 1 - n_e/n_a$ // Fraction of neutrals still available

if $D < 0$ **then**

  $D \leftarrow 0$ // Clamp to lower bound

end

if $D > 1$ **then**

  $D \leftarrow 1$ // Clamp to upper bound

end

3. ODE Coefficients — Free-Electron Channel n_e:

$S_e \leftarrow W_{avg}\, D$ // Valence photoionization source

if $E_{ste} \neq 0$ **then**

  $S_e \leftarrow S_e + (W_{avg}^s + \beta_s I_{avg} n_e)\,(n_s/n_a)$ // STE→CB re-ionization: photo + avalanche

end

$L_e \leftarrow \beta_g I_{avg} D - \tau_r^{-1}$ // Avalanche gain minus trapping loss

4. ODE Coefficients — STE Channel n_s:

if $E_{ste} \neq 0$ **then**

  $S_s \leftarrow \tau_r^{-1} n_e$ // Gain from free-electron trapping

  $L_s \leftarrow -(W_{avg}^s + \beta_s I_{avg} n_e)/n_a$ // Loss by STE re-ionization (photo + avalanche)

else

  $S_s \leftarrow 0$ // No STE source if disabled

  $L_s \leftarrow 0$ // No STE loss if disabled

end

5. Exact Exponential Integration:

$n_e^{new} \leftarrow \text{ExactExpStep}(n_e, S_e, L_e, \Delta t)$ // Advance free electrons

if $E_{ste} \neq 0$ **then**

  $n_s^{new} \leftarrow \text{ExactExpStep}(n_s, S_s, L_s, \Delta t)$ // Advance STEs

else

  $n_s^{new} \leftarrow 0$ // Force zero STE density

end

6. Density Conservation Constraint:

if $n_e^{new} + n_s^{new} > n_a$ **then**

  $\text{scale} \leftarrow n_a/(n_e^{new} + n_s^{new})$ // Rescaling factor

  $n_e^{new} \leftarrow n_e^{new} \times \text{scale}$ // Rescale electron density

  $n_s^{new} \leftarrow n_s^{new} \times \text{scale}$ // Rescale STE density

end

7. Commit & Register Update:

$n_e[base + j + 1] \leftarrow n_e^{new}$ // Write electrons to global memory

$n_s[base + j + 1] \leftarrow n_s^{new}$ // Write STEs to global memory

$n_e \leftarrow n_e^{new}$ // Update local n_e register

$n_s \leftarrow n_s^{new}$ // Update local n_s register

$I_c \leftarrow I_n$ // Update local intensity register

$W_c \leftarrow W_n,\ W_c^s \leftarrow W_n^s$ // Update local rate registers

Algorithm 7: Two-population plasma rate-equation solver (Part 2/2): sequential trapezoidal time-stepping per radial point.

Function BuildGrids *(Part 1/2: Grids & Propagation Operators)*:

Input: Configuration `cfg`, providing pump wavenumber k_0 (`komega`), carrier ω_0, and laser/material/grid parameters

Output: Intermediate grids and operators (assembled into `g` in Part 9)

1. Temporal & Spectral Grids:

$T_p \leftarrow$ `cfg.delta_t`$/\sqrt{2\ln 2}$ `// Intensity FWHM` $\rightarrow$ `field` $1/e$ `half-width`

$t_{max} \leftarrow 5\,T_p$ `// Window long enough for the pulse to vanish`

$\Delta t \leftarrow 2t_{max}/$`cfg.Nt` `// Uniform time step`

$t_{list} \leftarrow \mathrm{Linspace}(-t_{max}, t_{max},$ `cfg.Nt`; endpoint=false) `// Periodic time axis for FFT`

$f_{list} \leftarrow \mathrm{FFTfreq}($`cfg.Nt`$, \Delta t)$ `// Frequency axis`

2. Radial Grid (Quasi-Discrete Hankel Transform):

$\{j_m\}_{m=1}^{N} \leftarrow$ Zeros of Bessel J_0 `// QDHT support points (`j_N`: last zero)`

$R \leftarrow$ `cfg.R_factor` $\cdot$ `cfg.w0` `// Spatial box radius`

$Y_{mn} \leftarrow \dfrac{2}{j_N}\,\dfrac{J_0(j_m j_n/j_N)}{J_1(j_m)^2}$ `// QDHT transform matrix (symmetric, self-inverse)`

$r_m \leftarrow j_m R/j_N$ `// Radial coordinates (`$r>0$`)`

$\rho_m \leftarrow j_m/R$ `// Transverse spatial frequencies` $k_\perp$

3. Linear Dispersion Operator (Exact Sellmeier):

$\omega_{safe} \leftarrow \mathrm{Clip}\big(\omega_0 + 2\pi f_{list},\ \frac{2\pi c}{5\,\mu m},\ \frac{2\pi c}{0.18\,\mu m}\big)$ `// Restrict to` $[0.18, 5]\,\mu$`m; avoid Sellmeier poles`

$n(\omega) \leftarrow \sqrt{1 + \sum_i B_i\lambda^2/(\lambda^2 - L_i^2)}$ `// Exact Sellmeier index,` $\lambda = 2\pi c/\omega_{safe}$

$k(\omega) \leftarrow \omega\, n(\omega)/c$ `// Wavenumber`

$k_1 \leftarrow \dfrac{k(\omega_0{+}\delta\omega) - k(\omega_0{-}\delta\omega)}{2\,\delta\omega},\quad \delta\omega = \omega_0/1000$ `// Inverse group velocity via central difference`

$\Delta k(\omega) \leftarrow k(\omega) - k_0 - k_1\,\Omega,\quad \Omega = 2\pi f_{list}$ `// Residual dispersion in co-moving frame`

4. Nonlinear Operators (Self-Steepening & Raman):

$\hat{T} \leftarrow 1 + f_{list}/f_0$ `// Self-steepening operator (`$f_0 =$ `cfg.frequency)`

$R(t) \leftarrow \Theta(t)\,\Omega_R^2\,\tau_s\,e^{-t/\tau_d}\sin(t/\tau_s)$ `// Causal Raman response (Heaviside` Θ`,` $t>0$`)`

$R_f \leftarrow \mathrm{FFT}\big(\mathrm{IFFTShift}(R(t))\big)\,\Delta t$ `// Raman response in spectral domain`

5. 2D Space-Time Meshgrids:

$(tt, rr) \leftarrow \mathrm{Meshgrid}(t_{list}, r_{list})$ `//` (r,t) `grids for pointwise operators`

$(\cdot, \rho\rho) \leftarrow \mathrm{Meshgrid}(t_{list}, \rho_{list})$ `// Transverse-frequency grid`

Algorithm 8: Grid and operator construction (Part 1/2): temporal/radial grids, dispersion, nonlinear operators (`build_grids`).

Function BuildGrids *(Part 2/2: Plasma Constants, Absorber, Keldysh LUTs)*:

Input: `cfg` and Part 8 outputs (rr, R, k_0, ω_0)

Output: Dictionary `g` of precomputed grids, operators, plasma constants, and photoionization LUTs

1. Drude Plasma Constant & Gated Coefficients:

$\sigma \leftarrow \dfrac{k_0 e^2 \tau_c}{n_0^2 m_D\, \varepsilon_0\, \omega_0\, (1 + \omega_0^2 \tau_c^2)} \times 10^4$ `// Inverse Bremsstrahlung cross-section;` $\times 10^4$: `m`$^2 \to$`cm`^{2} (m_D = `cfg.meff_drude`)

if *cfg.enable_avalanche* **then**

  $\beta_g \leftarrow \sigma / U_i$ `// Valence avalanche coefficient`

else

  $\beta_g \leftarrow 0$ `// Avalanche disabled`

end

if *cfg.enable_ste* $\wedge$ *cfg.enable_avalanche* **then**

  $\beta_s \leftarrow \sigma / U_s$ `// STE re-ionization coefficient`

else

  $\beta_s \leftarrow 0$ `// STE avalanche disabled`

end

if *cfg.enable_recombination* **then**

  $\tau_r^{-1} \leftarrow 1/\tau_r$ `// Trapping/recombination rate`

else

  $\tau_r^{-1} \leftarrow 0$ `// Recombination disabled`

end

$\kappa \leftarrow \frac{1}{2} n_0 c\, \varepsilon_0 \times 10^{-4}$ `// Field-to-intensity factor (`$|\mathcal{E}|^2 \to$`W/cm`^{2}`)`

2. Boundary Absorber:

$\text{mask}_r \leftarrow \exp\big(-(rr/0.9R)^{20}\big)$ `// Super-Gaussian radial absorber`

3. Keldysh Photoionization LUTs (Dual Channel):

$I_{cpu} \leftarrow$ Log-spaced intensity grid $[1,\ \sim 10^{17}]$ W/cm^2 `// Spline sampling nodes`

foreach *gap* $(U, LUT) \in \{(U_i, LUT_g),\ (U_s, LUT_s)\}$ **do**

  $W(I) \leftarrow \max\big(\text{NaNtoNum}(\text{KeldyshRate}(\lambda, U, m_{eff}, n_0; I_{cpu})),\ 0\big)$ `// Evaluate & sanitize analytical rate`

  $f_{spline} \leftarrow \text{PCHIP}(I_{cpu}, W)$ `// Monotone continuous interpolant`

  $\log I \leftarrow \text{Linspace}(\log_{10} I_{min}, \log_{10} I_{max}, 4096)$ `// Uniform log-intensity LUT grid`

  $\text{LUT} \leftarrow \text{NaNtoNum}\big(|f_{spline}(10^{\log I})|\big)$ `// Sample onto GPU LUT array`

end

4. Return Grid Dictionary:

return `g` $\leftarrow \{t_{list}, r_{list}, Y, \Delta k, k_1, \hat{T}, R_f, rr, tt, \rho\rho, \sigma, \beta_g, \beta_s, \tau_r^{-1}, \kappa, \text{mask}_r, \text{LUT}_g, \text{LUT}_s, \dots\}$ `// Precomputed state (+ LUT params` $\log I_{min}, \Delta_{\log}^{-1}, I_{min}, I_{max}$`)`

Algorithm 9: Grid and operator construction (Part 2/2): plasma constants, boundary absorber, dual Keldysh LUTs (`build_grids`).

Function EnvelopeGaussianFocused *(Initial Gaussian Envelope)*:

Input: Radial grid r, temporal grid t, configuration `cfg`, precalculated parameters dictionary `g`

Output: Complex spatiotemporal electric field envelope array $E(r,t)$

1. Beam Parameters Definition:

$T_p \leftarrow \texttt{cfg.delta\_t}/\sqrt{2\ln(2)}$ `// Conversion between Intensity FWHM and E-field 1/e half-width`

$C \leftarrow 1 + i\frac{2\cdot\texttt{cfg.begin}}{\texttt{g["b"]}}$ `// Complex beam curvature factor at initial propagation distance`

2. Field Construction:

$E(r,t) \leftarrow \frac{\texttt{g["E0"]}}{C}\exp\left(-\frac{r^2}{\texttt{cfg.w0}^2 C}\right)\exp\left(-\frac{t^2}{T_p^2}\right)$ `// Gaussian transverse profile and temporal envelope coupling`

return $E(r,t)$ `// Initialized spatiotemporal envelope`

Algorithm 10: Focused Gaussian Beam Initializer (`envelope_gaussian_focused`)

Class LinearOperator *(Linear Propagation Steps)*:

Method Constructor *(__init__): Operator Initialization*:

Input: Configuration object `cfg`, precalculated parameters dictionary `g`

Store base grids and arrays $k(\omega)$, Y, $j_{1,\text{last}}$, R, ρ, Δk as instance attributes `// Model grids and operators caching`

$R_k \leftarrow R^2/j_{1,\text{last}}$ `// Forward QDHT normalization factor`

$R_k^{-1} \leftarrow j_{1,\text{last}}/R^2$ `// Inverse QDHT normalization factor`

Method HalfDiffraction *(half_diffraction): Transverse Spatial Propagation*:

Input: Complex field envelope $u(r,t)$, propagation step Δz

Output: Diffracted field envelope $u(r,t)$

1. Forward Hankel Transform:

$\tilde{u}(\rho,t) \leftarrow R_k \cdot (Y \cdot u)$ `// Transform to radial spatial frequency domain (QDHT)`

2. Diffraction Phase Shift:

$\tilde{u}(\rho,t) \leftarrow \tilde{u}(\rho,t) \cdot \exp\left(-i\frac{\rho^2}{2k(\omega)}\frac{\Delta z}{2}\right)$ `// Apply paraxial half-step diffraction operator`

3. Inverse Hankel Transform:

$u(r,t) \leftarrow R_k^{-1} \cdot (Y \cdot \tilde{u})$ `// Return to spatial domain`

return $u(r,t)$

Method HalfDispersion *(half_dispersion): Longitudinal Temporal Propagation*:

Input: Complex field envelope $u(r,t)$, propagation step Δz

Output: Dispersed field envelope $u(r,t)$

1. Forward Fourier Transform:

$\tilde{u}(r,\omega) \leftarrow \text{FFT}(u)$ `// Transform to spectral domain along temporal axis`

2. Dispersion Phase Shift:

$\tilde{u}(r,\omega) \leftarrow \tilde{u}(r,\omega) \cdot \exp\left(i\Delta k\frac{\Delta z}{2}\right)$ `// Apply chromatic dispersion half-step operator`

3. Inverse Fourier Transform:

$u(r,t) \leftarrow \text{iFFT}(\tilde{u})$ `// Return to time domain`

return $u(r,t)$

Algorithm 11: Linear Propagation Operators (`LinearOperator`)

Class NONLINEAROPERATOR *(Part 1/2: Setup & Matter Dynamics)*:

Method CONSTRUCTOR *(__init__): Operator Initialization*:

Input: Configuration object `cfg`, precalculated parameters dictionary `g`

Store scalar constants n_0, U_i, f_R, grids $\hat{T}, \tilde{R}_f$ and Keldysh LUTs as instance attributes `// Model grids and physical parameters caching`

1. Nonlinear Optical Coefficients:

if $cfg.enable\_kerr \neq 0$ **then**

$\kappa_{\text{Kerr}} \leftarrow \frac{3 \cdot \texttt{cfg.chi3} \cdot \texttt{cfg.omega0}^2}{8 \cdot k(\omega) \cdot c^2}$ `// Derive Kerr self-focusing nonlinear coefficient`

else

$\kappa_{\text{Kerr}} \leftarrow 0.0$ `// Kerr effect computationally disabled`

end

2. Drude Plasma Coefficients:

$\kappa_{\text{pl}} \leftarrow \frac{\sigma(\omega)}{2} \cdot 100.0$ `// Inverse Bremsstrahlung absorption prefactor`

$C_{\text{pl}} \leftarrow 1 + i \cdot \texttt{cfg.omega0} \cdot \texttt{cfg.tau\_c}$ `// Complex Drude model plasma phase and absorption factor`

Method UPDATEPLASMA *(update_plasma): Electron Density Integration*:

Input: Field envelope u, densities ρ, ρ_s, time step Δt

Output: Updated plasma densities ρ, ρ_s (in-place modification)

1. Boundary Conditions:

$\rho(t = t_0) \leftarrow 0$ `// Initial free electron density at the front of the pulse`

$\rho_s(t = t_0) \leftarrow 0$ `// Initial Self-Trapped Exciton density at the front of the pulse`

2. Rate Equation Integration:

SOLVEPLASMARATEEQUATIONS$(u, \rho, \rho_s, \Delta t, \text{LUTs})$ `// Abstract CUDA kernel evaluating MPI, avalanche and STE dynamics`

return `// Densities are updated directly in device memory`

Algorithm 12: Nonlinear Operator: Initialization & Plasma Rate Equations

Class NonlinearOperator *(Part 2/2: Field Evolution RHS)*:

Method NonlinearRHS *(`split`): Evaluate Source Terms*:

Input: Complex field envelope $u(r,t)$, free electron density $\rho(r,t)$

Output: Tuple containing RHS derivative $\partial_z u$ and absorption map α

1. Photoionization and Absorption:

$I(r,t) \leftarrow |u|^2$ `// Calculate instantaneous spatiotemporal intensity profile`

$W_{\mathrm{PI}} \leftarrow \mathrm{EvaluateSpline}(I \cdot E_{\mathrm{ref}}^{-2})$ `// Calculate photoionization rate from Keldysh lookup table`

$D \leftarrow \mathrm{Clip}\left(1 - \frac{\rho}{\rho_{\max}}, 0, 1\right)$ `// Calculate bounded valence band neutral depletion factor`

$\alpha_{\mathrm{PI}} \leftarrow \left(W_{\mathrm{PI}} \cdot 10^6\right) \frac{U_i}{n_0 c \varepsilon_0 (I + 10^{-30})} \cdot D$ `// Derive nonlinear multiphoton absorption coefficient`

$\alpha_{\mathrm{total}} \leftarrow \mathrm{Max}(\kappa_{\mathrm{pl}}\rho + \alpha_{\mathrm{PI}}, 0)$ `// Combine plasma heating and MPI spatial absorption profile`

2. Kerr and Raman Delayed Response:

$I_{\mathrm{Kerr}} \leftarrow (1 - f_R) \cdot I + f_R \cdot \mathrm{Re}\left\{\mathrm{iFFT}(\mathrm{FFT}(I) \cdot \tilde{R}_f)\right\}$ `// Convolve intensity with delayed Raman molecular response`

3. Field Evolution (RHS):

$\tilde{N}_\omega \leftarrow \mathrm{FFT}\left(i\kappa_{\mathrm{Kerr}} I_{\mathrm{Kerr}} u - \alpha_{\mathrm{PI}} u\right)$ `// Compute nonlinear polarization source term in spectral domain`

$\mathrm{RHS} \leftarrow \mathrm{iFFT}(\tilde{N}_\omega \cdot \hat{T}) + \alpha_{\mathrm{PI}} u - \kappa_{\mathrm{pl}}(C_{\mathrm{pl}} - 1)\rho u$ `// Apply self-steepening operator and Drude plasma defocusing`

4. Vacuum Noise Suppression:

foreach *grid point where* $I < 10^{-30}$ **do**

  $\mathrm{RHS} \leftarrow 0 + 0i$ `// Force strict zero derivative in vanishing field regions`

end

return $\{\mathrm{RHS}, \alpha_{\mathrm{total}}\}$ `// Provide nonlinear derivative and combined diagnostic absorption`

Algorithm 13: Nonlinear Operator: Spatiotemporal Nonlinear Source Terms

Class INTEGRATOR *(Part 1/2: Setup & Main Loop)*:

Method CONSTRUCTOR *(`__init__`): Engine Initialization*:

Input: Configuration object `cfg`, Envelope definition `envelope`

1. Physical Grids & Operators:

$g \leftarrow$ BUILDGRIDS(`cfg`) `// Generate spatial, temporal and spectral discretized grids`

$\Delta z, \Delta t \leftarrow$ `cfg.dz`, g["dt"] `// Extract longitudinal and temporal integration steps`

$\hat{L} \leftarrow$ LINEAROPERATOR(`cfg`, g) `// Instantiate diffraction and dispersion solver`

$\hat{N} \leftarrow$ NONLINEAROPERATOR(`cfg`, g) `// Instantiate nonlinear polarization and plasma solver`

2. Fields & Energy Initialization:

$u(r,t) \leftarrow$ ENVELOPEINITIALIZER($r, t,$ `cfg`, g) `// Evaluate complex spatiotemporal initial envelope`

$\rho(r,t), \rho_s(r,t) \leftarrow 0, 0$ `// Initialize free electron and STE densities to zero`

$F_0(r) \leftarrow \sum_t |u(r,t)|^2 \cdot E_{\mathrm{ref}}^{-2} \cdot \Delta t \cdot 10^4$ `// Integrate initial temporal intensity to get radial fluence`

$U_0 \leftarrow \sum_r F_0(r) \cdot 2\pi r \Delta r \cdot 10^6$ `// Integrate radial fluence for total initial pulse energy (`μ`J)`

3. Diagnostics History Arrays Allocation:

$N_{\mathrm{save}} \leftarrow$ Floor(`cfg.nz`/`cfg.save_stride`) $+ 1$ `// Number of longitudinal checkpoints`

Allocate memory for $F(z,r), \rho_{\max}(z,r), I_{\mathrm{peak}}(z), E_{\mathrm{dissipated}}(z)$ `// Initialize history arrays for macroscopic observables`

if *`cfg.rho_t_stride`* > 0 **then**

  Allocate subsampled arrays $\rho(z,r,t_{\mathrm{sub}}), I(z,r,t_{\mathrm{sub}})$ `// Initialize history arrays for spatio-temporal dynamics`

end

Method PROPAGATE *(`propagate`): Main Evolution Loop*:

Input: Configuration object `cfg`

Output: Dictionary of simulated physical diagnostics

foreach *step index* $i \in \{0, \ldots,$ *`cfg.nz`*$\}$ **do**

  1. Matter Dynamics Integration:

  $\hat{N}$.UPDATEPLASMA($u, \rho, \rho_s, \Delta t$) `// Integrate electron density temporally at current` z

  2. Optical Field Propagation:

  PROPAGATIONSTEP(Δz) `// Advance optical envelope spatially by` Δz

  $z \leftarrow$ `cfg.begin` $+ i \cdot \Delta z$ `// Update physical propagation distance`

  3. Diagnostics Checkpoint:

  if i mod *`cfg.save_stride`* $== 0$ **then**

    RECORDSTATE(z) `// Sample fluences, peak values, and densities`

  end

end

return FORMATRESULTS() `// Post-process fields and unfold symmetries for export`

Algorithm 14: Split-Step Fourier Integrator: Initialization & Main Loop

Class Integrator *(Part 2/2: Stepping Engine & Diagnostics)*:

Method PropagationStep *(`step`): Single Z-Step Advance*:

Input: Current field $u(r,t)$, plasma density $\rho(r,t)$, spatial step size Δz
Output: Updated field $u(r,t)$ evaluated at $z + \Delta z$ (in-place)

1. Linear Pre-Compensation:

$u \leftarrow \hat{L}.\text{HalfDiffraction}(u, \Delta z)$ `// Apply half-step paraxial diffraction (QDHT domain)`

$u \leftarrow \hat{L}.\text{HalfDispersion}(u, \Delta z)$ `// Apply half-step chromatic dispersion (FFT domain)`

2. Nonlinear Source Runge-Kutta 4 Integration:

$\_, \alpha \leftarrow \hat{N}.\text{NonlinearRHS}(u, \rho)$ `// Evaluate initial nonlinear multiphoton and plasma absorption`

$u \leftarrow u \cdot \exp\left(-\frac{\Delta z}{2}\alpha\right)$ `// Apply pre-RK4 symmetric absorption half-step`

$k_1, \_ \leftarrow \hat{N}.\text{NonlinearRHS}(u, \rho)$ `// RK4 Stage 1: Initial field derivative`

$k_2, \_ \leftarrow \hat{N}.\text{NonlinearRHS}(u + \frac{\Delta z}{2}k_1, \rho)$ `// RK4 Stage 2: Midpoint derivative prediction`

$k_3, \_ \leftarrow \hat{N}.\text{NonlinearRHS}(u + \frac{\Delta z}{2}k_2, \rho)$ `// RK4 Stage 3: Midpoint derivative correction`

$k_4, \_ \leftarrow \hat{N}.\text{NonlinearRHS}(u + \Delta z \cdot k_3, \rho)$ `// RK4 Stage 4: Endpoint derivative`

$u \leftarrow u + \frac{\Delta z}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ `// Advance complex field using weighted nonlinear source terms`

$\_, \alpha \leftarrow \hat{N}.\text{NonlinearRHS}(u, \rho)$ `// Evaluate final nonlinear absorption after RK4 update`

$u \leftarrow u \cdot \exp\left(-\frac{\Delta z}{2}\alpha\right)$ `// Apply post-RK4 symmetric absorption half-step`

3. Linear Post-Compensation:

$u \leftarrow \hat{L}.\text{HalfDispersion}(u, \Delta z)$ `// Apply half-step chromatic dispersion`

$u \leftarrow \hat{L}.\text{HalfDiffraction}(u, \Delta z)$ `// Apply half-step paraxial diffraction`

$u \leftarrow u \cdot \text{mask}_r$ `// Apply absorbing boundary condition at radial edge`

Method RecordState *(`_record`): Observables Sampling*:

Input: Current longitudinal position z

$I(r,t) \leftarrow |u(r,t)|^2 \cdot E_{\text{ref}}^{-2}$ `// Convert internal complex field to physical intensity`

1. Macroscopic Peak & Integrated Observables:

$F_{\text{history}}(z_k, r) \leftarrow \sum_t I(r,t) \cdot \Delta t$ `// Calculate and store temporal fluence profile`

$\rho_{\text{history}}(z_k, r) \leftarrow \max_t \rho(r,t)$ `// Extract and store peak free electron density`

$I_{\max}(z_k) \leftarrow \max_{r,t} I(r,t)$ `// Extract absolute maximum intensity on grid`

2. Spatio-temporal Sampling:

if $cfg.rho\_t\_stride > 0$ **then**

  $t_{\text{sub}} \leftarrow t$[every stride steps] `// Extract decimated temporal grid indices`

  $\rho_{\text{history\_3D}}(z_k, r, t_{\text{sub}}) \leftarrow \rho(r, t_{\text{sub}})$ `// Store time-resolved plasma density profile`

  $I_{\text{history\_3D}}(z_k, r, t_{\text{sub}}) \leftarrow I(r, t_{\text{sub}})$ `// Store time-resolved intensity profile`

end

$k \leftarrow k + 1$ `// Increment global checkpoint inde`

Function RunSimulation *(`run`): Simulation Entry Point*:

Input: Laser parameters $(\lambda, w_0, \Delta t, E_{\mu J})$, material properties (n_2, U_i), numerical grid sizes (N_z, N_r, N_t), and integration boundaries $(z_{\text{begin}}, z_{\text{end}})$

Output: Dictionary of simulated macroscopic physical observables

1. State Configuration:

`cfg` ← Config($N_z, N_t, N_r, \lambda, E_{\mu J}, w_0, \Delta t, z_{\text{begin}}, z_{\text{end}}, n_2, U_i, \ldots$)

`// Aggregate physical and numerical properties into global state`

2. Engine Instantiation & Execution:

solver ← Integrator(`cfg`, `envelope`)

`// Initialize physical grids and split-step Fourier propagator`

results ← solver.Propagate()

`// Execute spatial propagation and aggregate diagnostics`

return results

`// Return full history of converged fields`

Algorithm 16: Global Simulation Orchestrator (`run`)